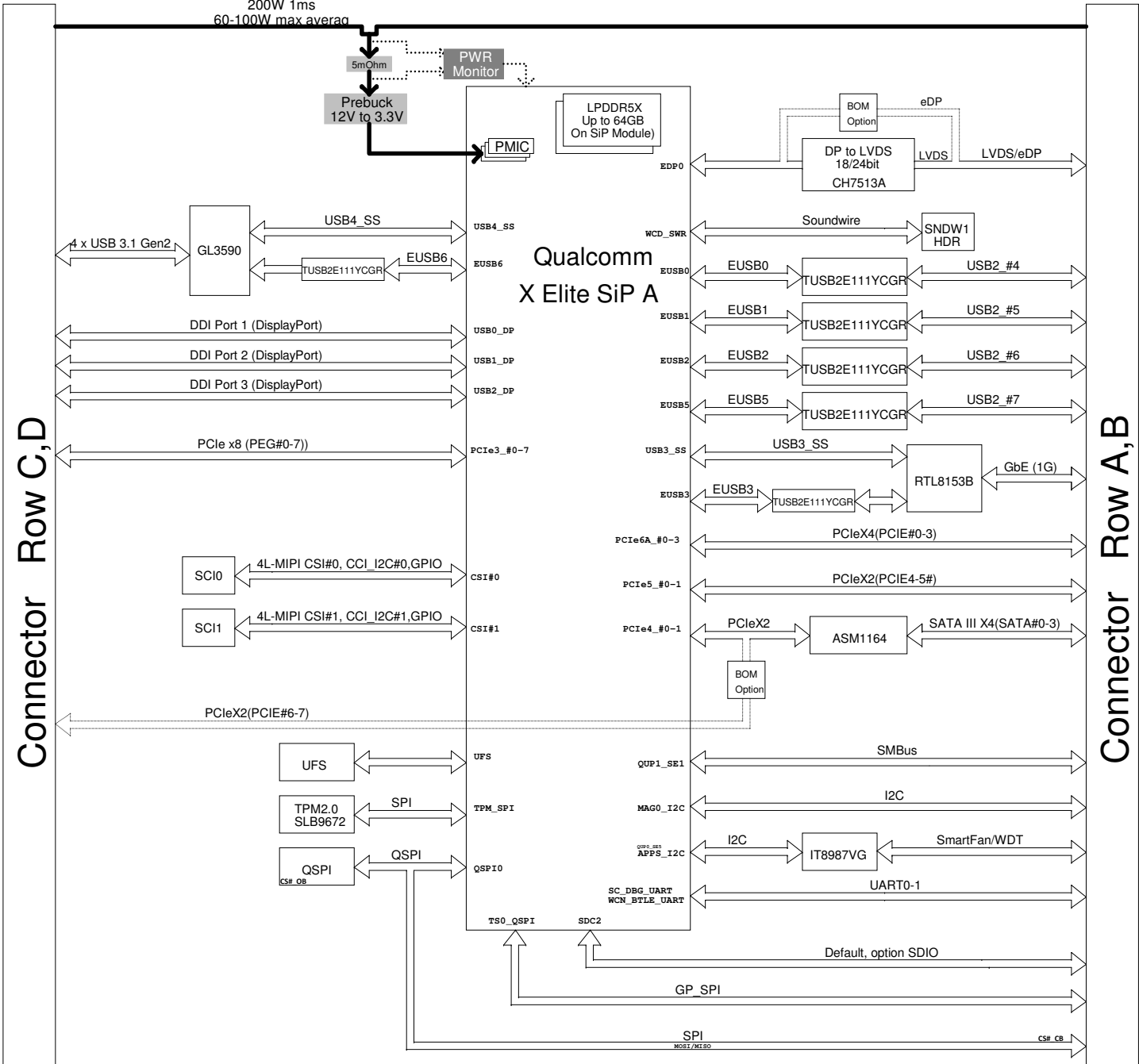


Page Name	Source Schematic	Source File	Source Page	Source Rev
PG01: Cover Sheet	SOM-6820_A1	SOM-6820_A101-2_20250505	PG01: Cover Sheet	01-2
PG02: Block Diagram	SOM-6820_A1	SOM-6820_A101-2_20250505	PG02: Block Diagram	01-2
PG03: Power Delivery Map	SOM-6820_A1	SOM-6820_A101-2_20250505	PG03: Power Delivery Map	01-2
PG04: Power Sequence	SOM-6820_A1	SOM-6820_A101-2_20250505	PG04: Power Sequence	01-2
PG05: X ELITE SiP_1-200	SOM-6820_A1	SOM-6820_A101-2_20250505	PG05: X ELITE SiP_1-200	01-2
PG06: X ELITE SiP_201-400	SOM-6820_A1	SOM-6820_A101-2_20250505	PG06: X ELITE SiP_201-400	01-2
PG07: X ELITE SiP_401-600	SOM-6820_A1	SOM-6820_A101-2_20250505	PG07: X ELITE SiP_401-600	01-2
PG08: X ELITE SiP_601-800	SOM-6820_A1	SOM-6820_A101-2_20250505	PG08: X ELITE SiP_601-800	01-2
PG09: X ELITE SiP_801-1000	SOM-6820_A1	SOM-6820_A101-2_20250505	PG09: X ELITE SiP_801-1000	01-2
PG10: X ELITE SiP_1001-1200	SOM-6820_A1	SOM-6820_A101-2_20250505	PG10: X ELITE SiP_1001-1200	01-2
PG11: X ELITE SiP_1201-1400	SOM-6820_A1	SOM-6820_A101-2_20250505	PG11: X ELITE SiP_1201-1400	01-2
PG12: X ELITE SiP_1401-1611	SOM-6820_A1	SOM-6820_A101-2_20250505	PG12: X ELITE SiP_1401-1611	01-2
PG13: CLKGEN_9FGL08	SOM-6820_A1	SOM-6820_A101-2_20250505	PG13: CLKGEN_9FGL08	01-2
PG14: USB3 HUB_GL3590	SOM-6820_A1	SOM-6820_A101-2_20250505	PG14: USB3 HUB_GL3590	01-2
PG15: eUSB TO USB2 REPEATER	SOM-6820_A1	SOM-6820_A101-2_20250505	PG15: eUSB TO USB2 REPEATER	01-2
PG16: SATA Bridge (ASM1164)	SOM-6820_A1	SOM-6820_A101-2_20250505	PG16: SATA Bridge (ASM1164)	01-2
PG17: eDP to LVDS - CH7513	SOM-6820_A1	SOM-6820_A101-2_20250505	PG17: eDP to LVDS - CH7513	01-2
PG18: LAN_RTL8153BI	SOM-6820_A1	SOM-6820_A101-2_20250505	PG18: LAN_RTL8153BI	01-2
PG19: TPM - SLB9672A	SOM-6820_A1	SOM-6820_A101-2_20250505	PG19: TPM - SLB9672A	01-2
PG20: QSPI NOR,UFS	SOM-6820_A1	SOM-6820_A101-2_20250505	PG20: QSPI NOR,UFS	01-2
PG21: IT8987	SOM-6820_A1	SOM-6820_A101-2_20250505	PG21: IT8987	01-2
PG22: Boot Config,Level Shfit	SOM-6820_A1	SOM-6820_A101-2_20250505	PG22: Boot Config,Level Shfit	01-2
PG23: COMe R3.1 Type6 RAW A/B	SOM-6820_A1	SOM-6820_A101-2_20250505	PG23: COMe R3.1 Type6 RAW A/B	01-2
PG24: COMe R3.1 Type6 RAW C/D	SOM-6820_A1	SOM-6820_A101-2_20250505	PG24: COMe R3.1 Type6 RAW C/D	01-2
PG25: DCIN,+V3P3A	SOM-6820_A1	SOM-6820_A101-2_20250505	PG25: DCIN,+V3P3A	01-2
PG26: +V1P8_DUAL	SOM-6820_A1	SOM-6820_A101-2_20250505	PG26: +V1P8_DUAL	01-2
PG27: +VPH_PWR	SOM-6820_A1	SOM-6820_A101-2_20250505	PG27: +VPH_PWR	01-2
PG28: +V3P3S,+V5S	SOM-6820_A1	SOM-6820_A101-2_20250505	PG28: +V3P3S,+V5S	01-2
PG29: +V1P2D,+V1P05D	SOM-6820_A1	SOM-6820_A101-2_20250505	PG29: +V1P2D,+V1P05D	01-2
PG30: MIPI-CSI HDR	SOM-6820_A1	SOM-6820_A101-2_20250505	PG30: MIPI-CSI HDR	01-2
PG31: Revision History	SOM-6820_A1	SOM-6820_A101-2_20250505	PG31: Revision History	01-2

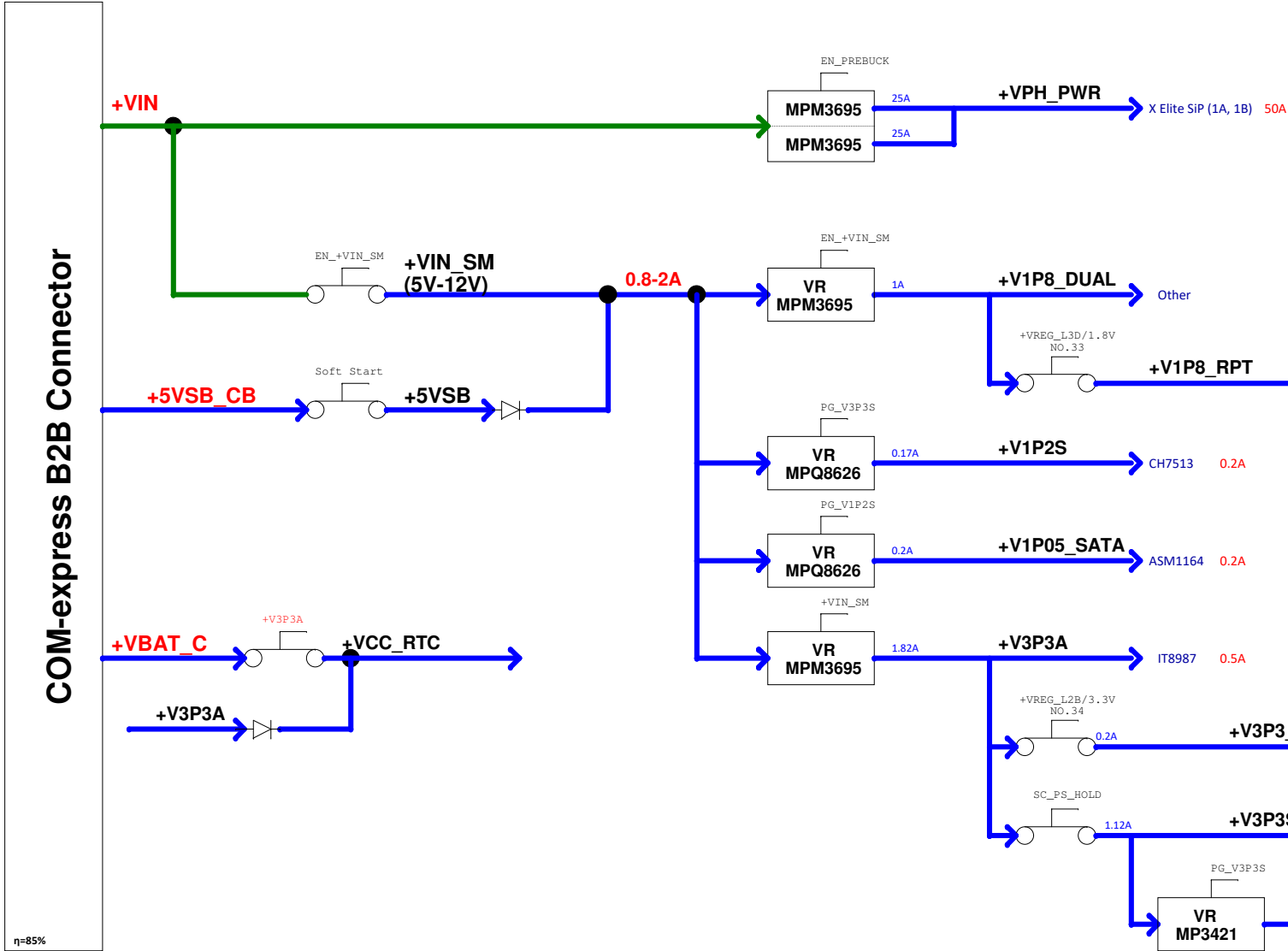
ADVANTECH		CONFIDENTIAL		CONFIDENTIAL DRAWING OWNED BY ADVANTECH. THE RECIPIENT SHALL MAINTAIN THE CONFIDENTIALITY AND SHALL BE LIABLE FOR BREACH.
Title PG01: Cover Sheet				
Size B	PCB Name SOM-6820_A1			Rev 01-2
Date:	Tuesday, May 20, 2025	Sheet	1	of 31

Block Diagram

COM-Express R3.1 Type 6

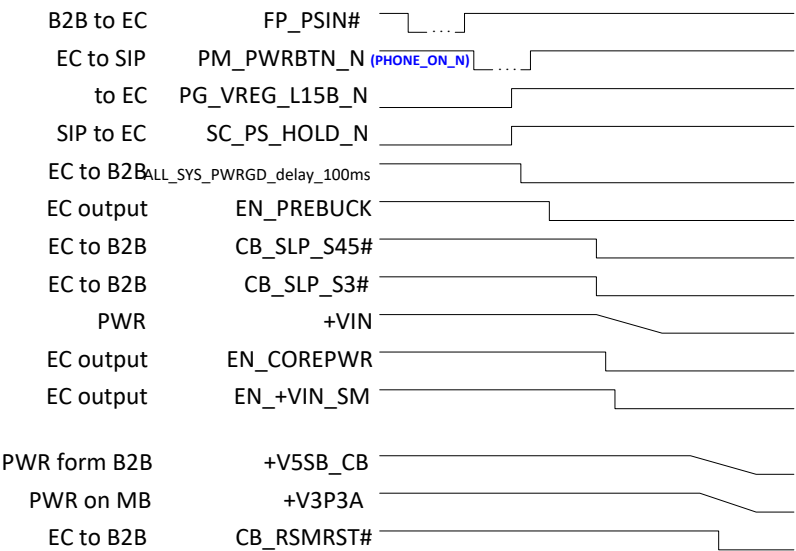
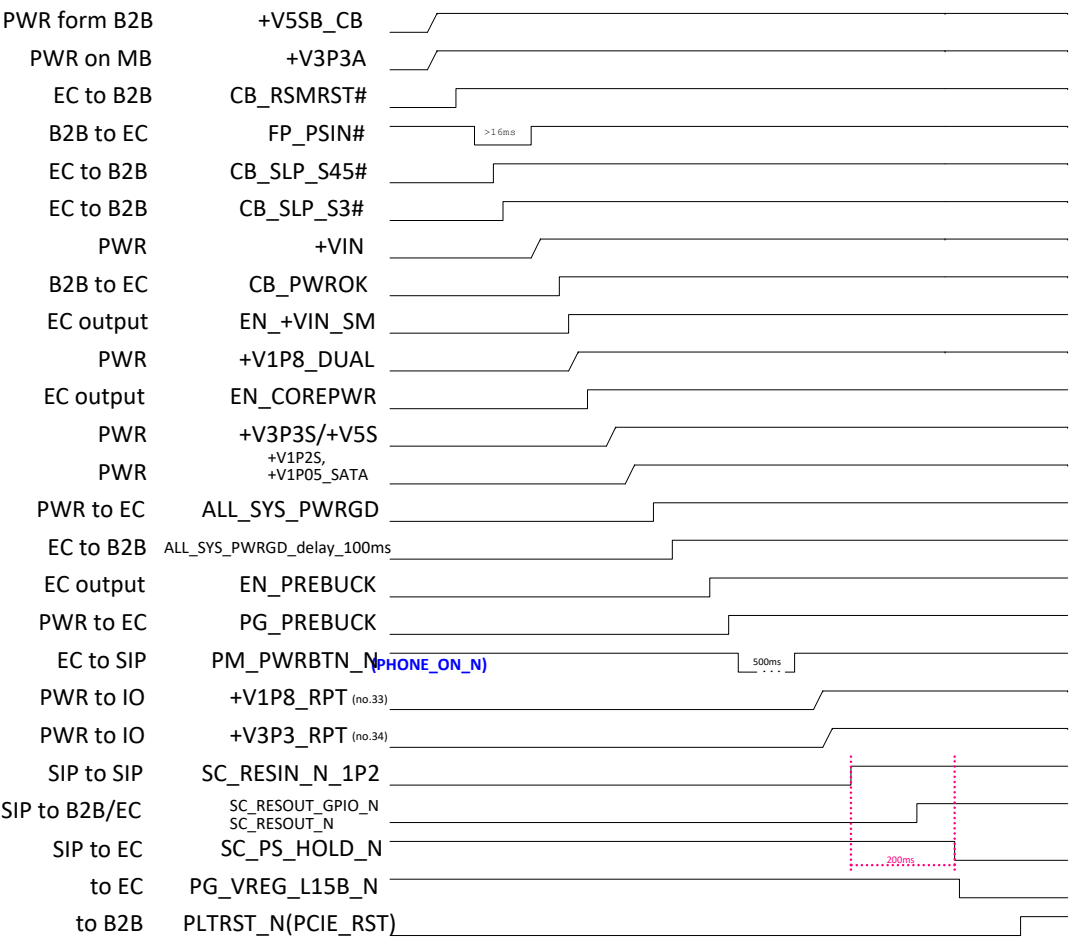


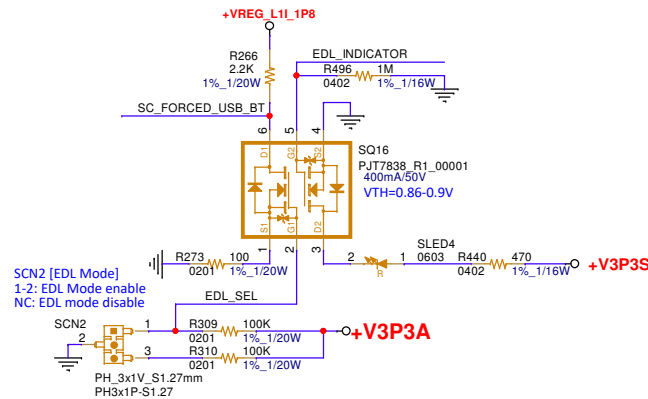
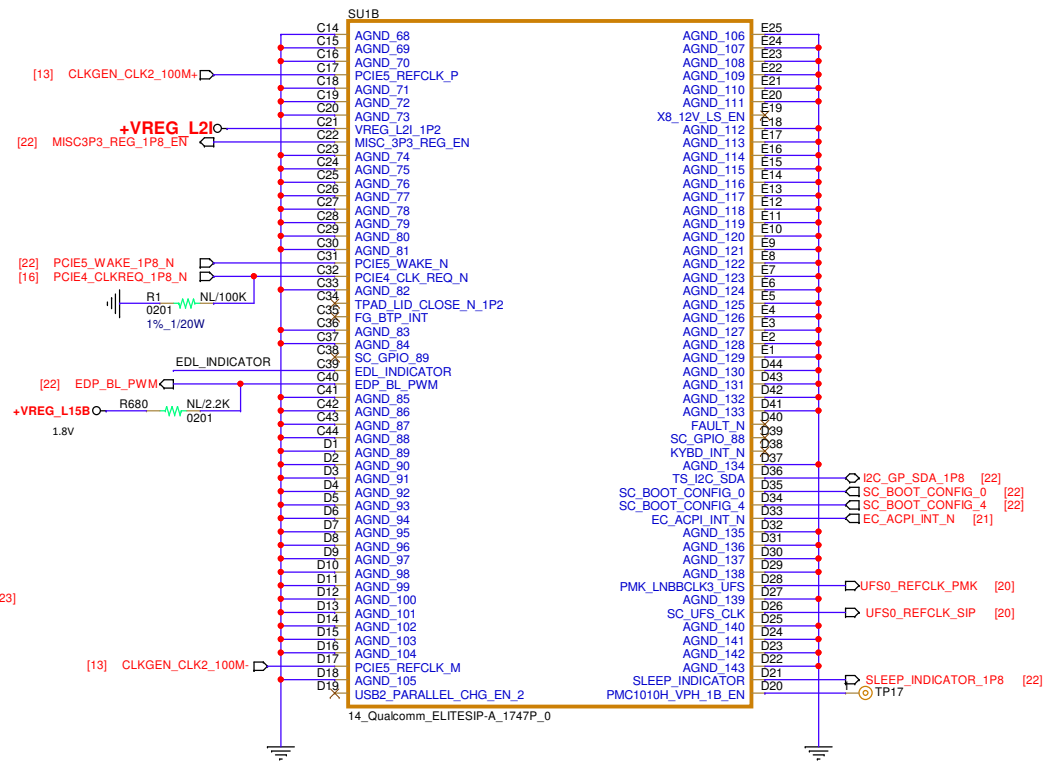
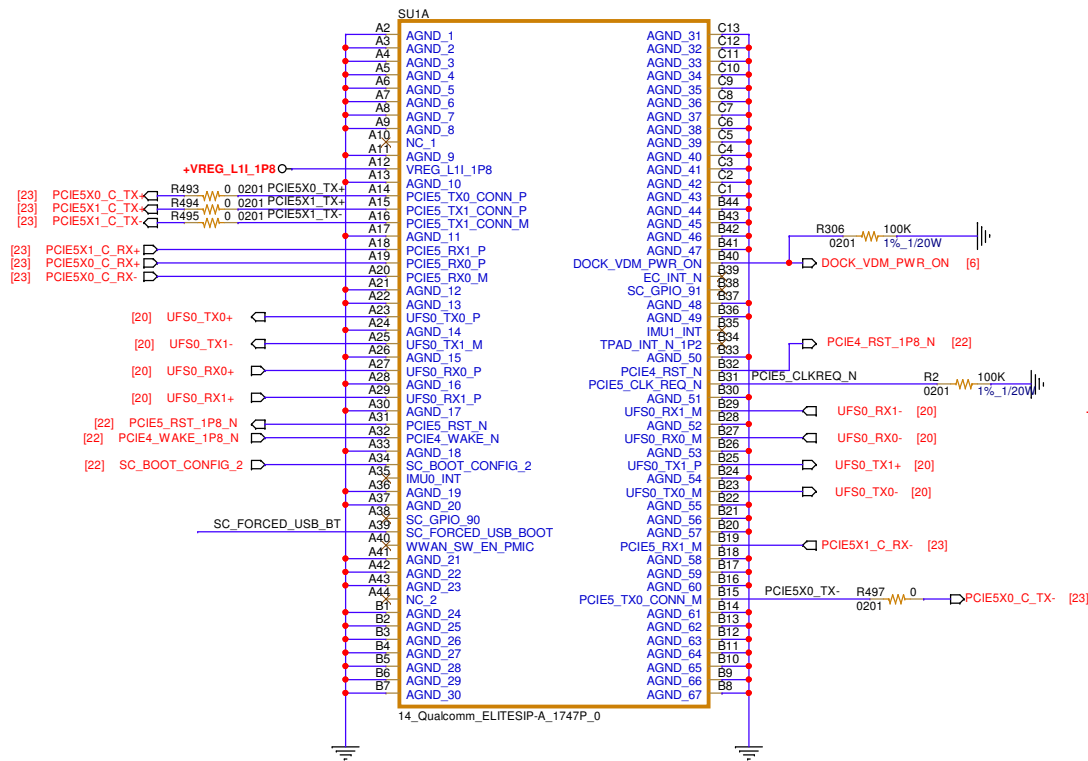
COM-express B2B Connector

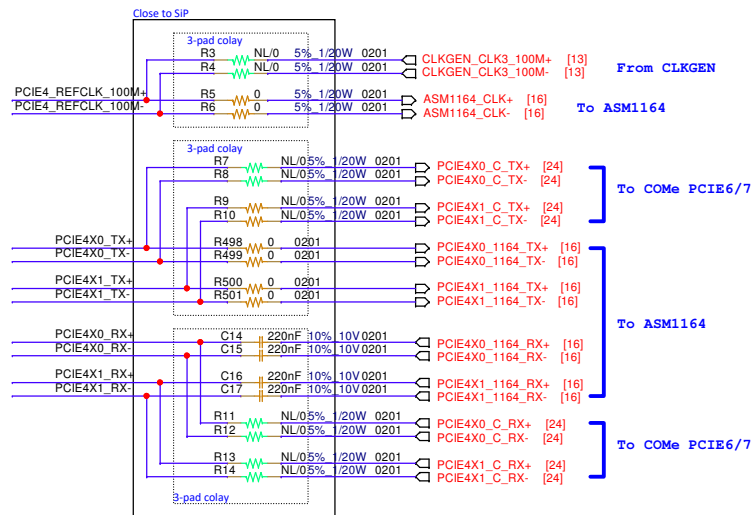
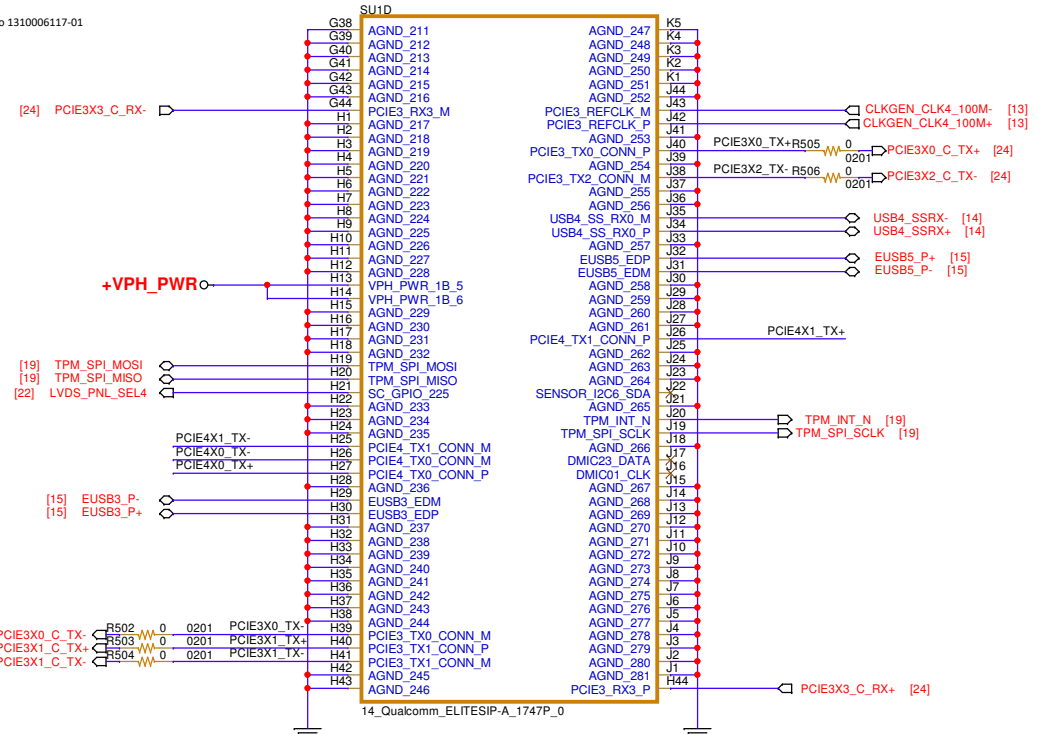
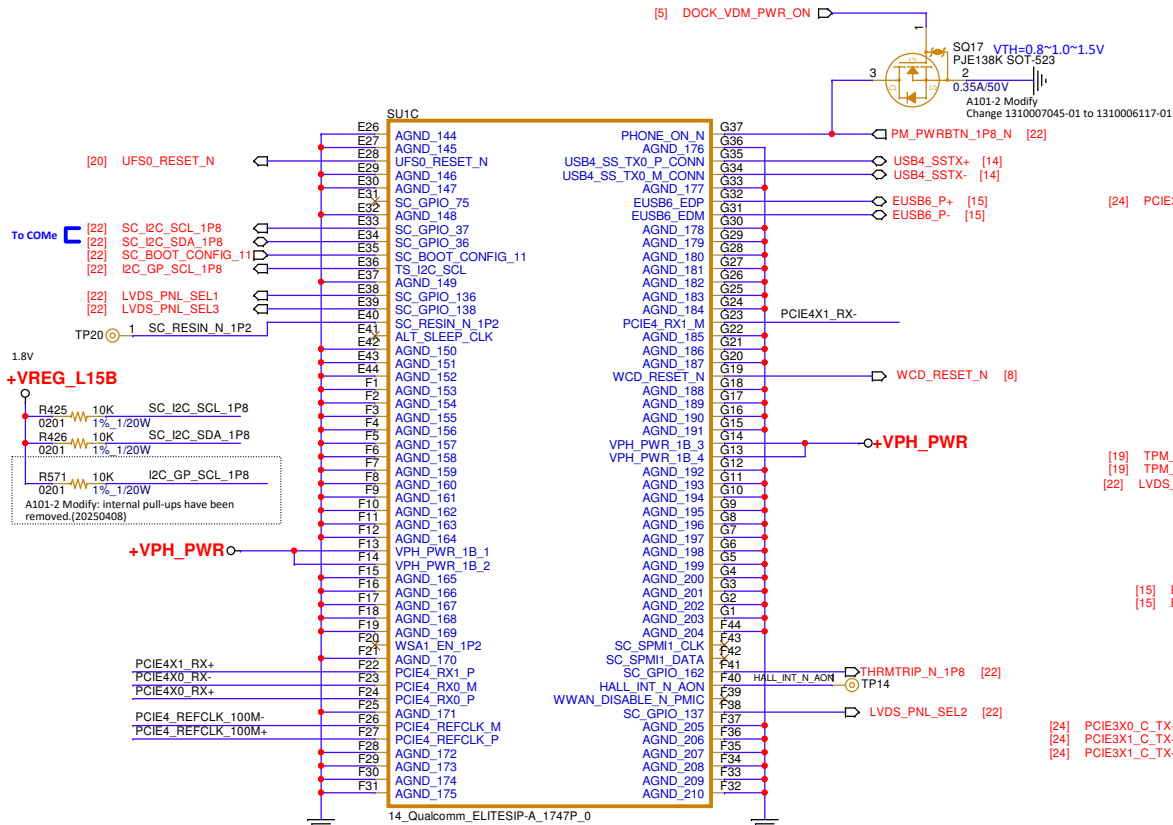


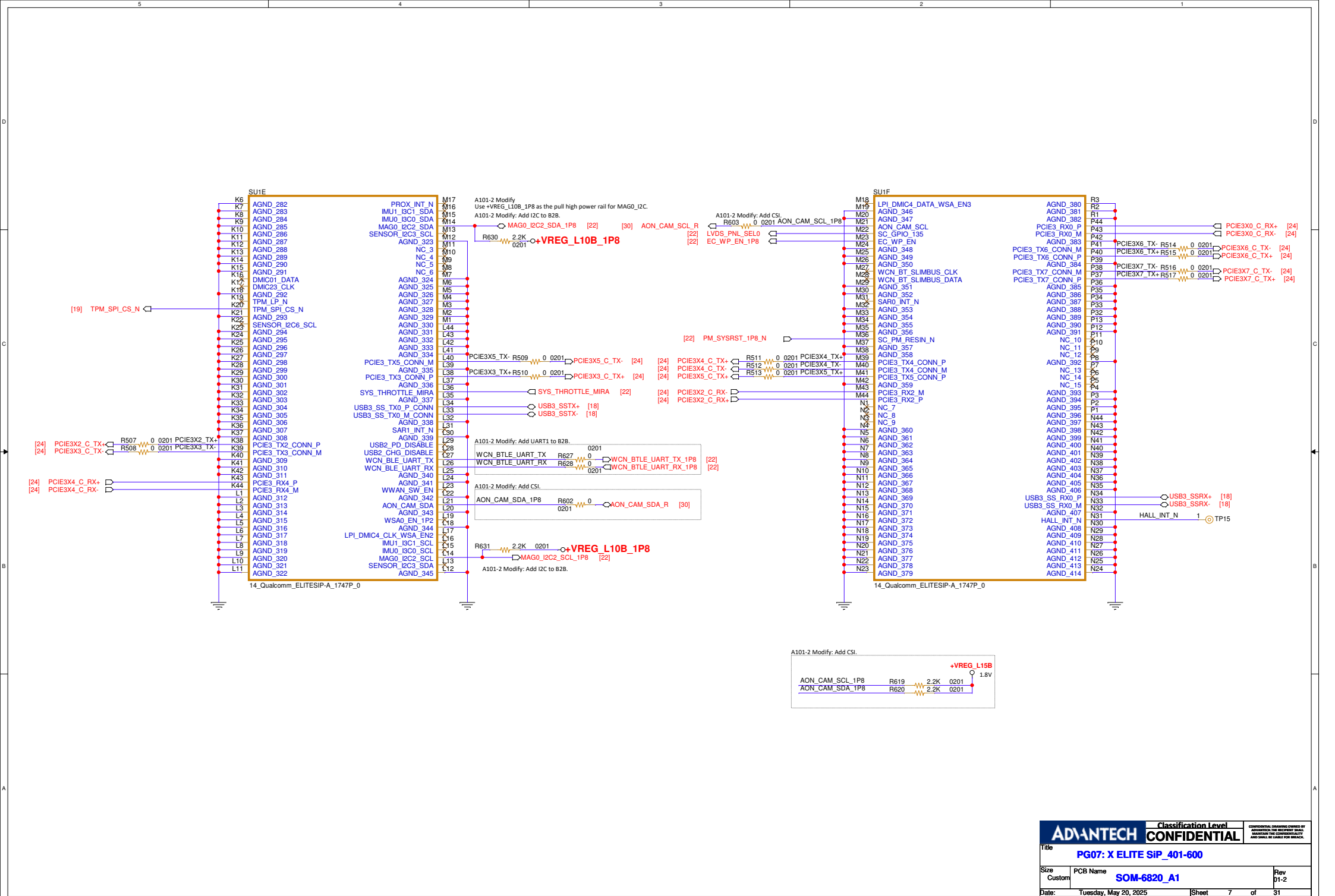
ADANTECH		Classification Level CONFIDENTIAL	CONFIDENTIAL: Maximum disclosure is restricted to those personnel who have a valid need to know. It is to be controlled and handled accordingly.
Title PG03: Power Delivery Map			
Size C	PCB Name SOM-6820_A1		Rev D1-2
Date: Tuesday, May 20, 2025		Sheet 3	of 31

Power Sequence









A101-2 Modify: Add CSI group 0 and 1 to header.

A101-2 Modify: Add GP_SPI to CB.

QUP1_SE2

[20] TS0_QSPI_CLK_1P8
[20] TS0_QSPI_MISO_1P8
[20] TS0_QSPI_MOSI_1P8

[20] QSPI0_CS1_N
[30] CSIO_RX3+
[30] CSIO_RX3-

A101-2 Modify: Add CSI.

A101-2 Modify: Add GP_SPI to CB.

QUP1_SE2

[20] TS0_QSPI_CS1_1P8

A101-2 Modify: Add CSI.

CAM0_PWR#_EN_1P8 R609 0 0201 CAM0_PWR#_EN_CN [30]
CAM0_RESET#_1P8 R610 0 0201 CAM0_RESET#_CN [30]
CAM0_MCK_1P8 R607 0 0201 CAM0_MCK_CN [30]

[22] SC_BOOT_CONFIG_1

CAM4_RESET#_1P8
A101-2 Modify: Add CSI.
[30] CSIO_RX0+
[30] CSIO_RX0-

A101-2 Modify: CH7513 PWRDN control by SIP GPIO_105 instead EC(Follow ED92 design)

1.8V +VREG_L15B R629 0 0201

[22] CH7513_PWRDN_1P8

CC11_I2C_SDA_1P8

[22] EDP0_HDP_DETECT

R586 1M 0201 1% 1/20W

[30] CSIO_CLK+
[30] CSIO_CLK-

A101-2 Modify: Add CSI.

CC11_I2C_SCL_1P8

A101-2 Modify: Add CSI.

CC11_I2C_SCL_1P8 R617 2.2K 0201

CC11_I2C_SDA_1P8 R618 2.2K 0201

CAM1_PWR#_EN_1P8 R611 0 0201

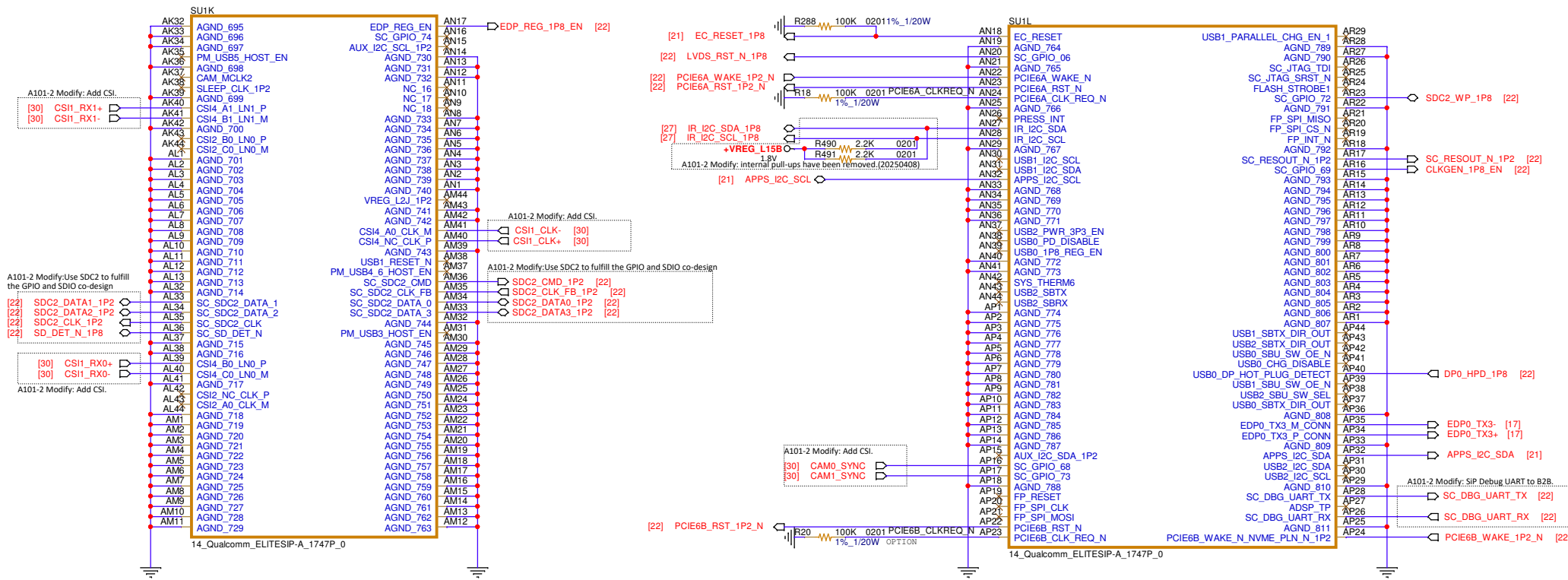
CAM4_RESET#_1P8 R612 0 0201

CAM1_PWR#_EN_CN [30]
CAM1_RESET#_CN [30]

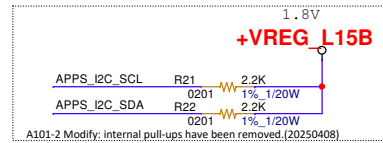
CAM_AON_MCLK4_1P8 R606 0 0201

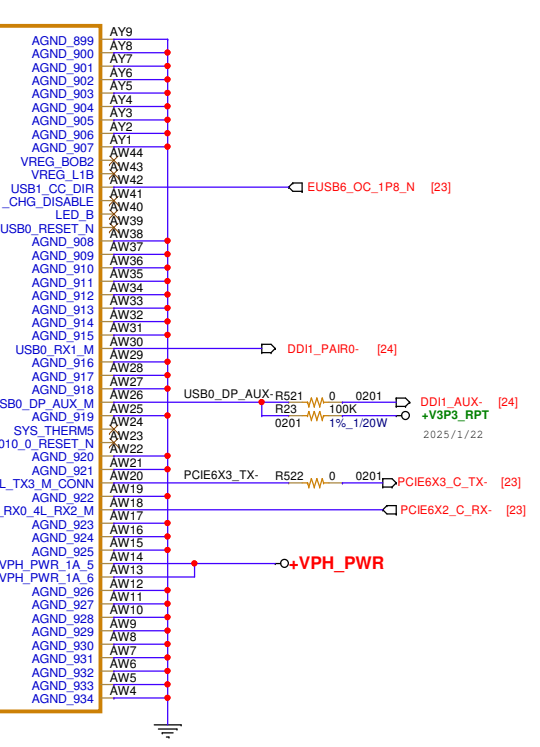
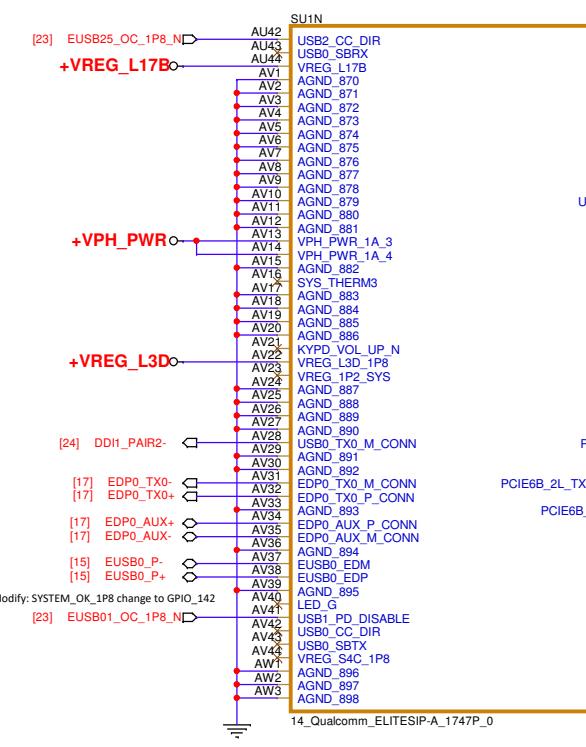
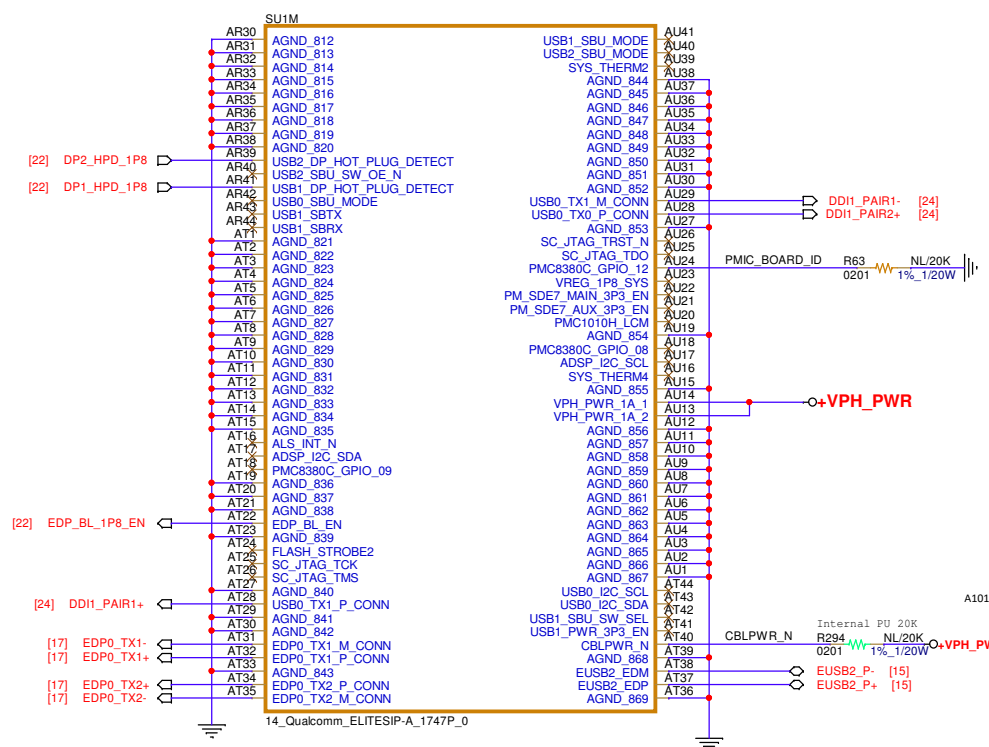
CAM_AON_MCLK4_R [30]

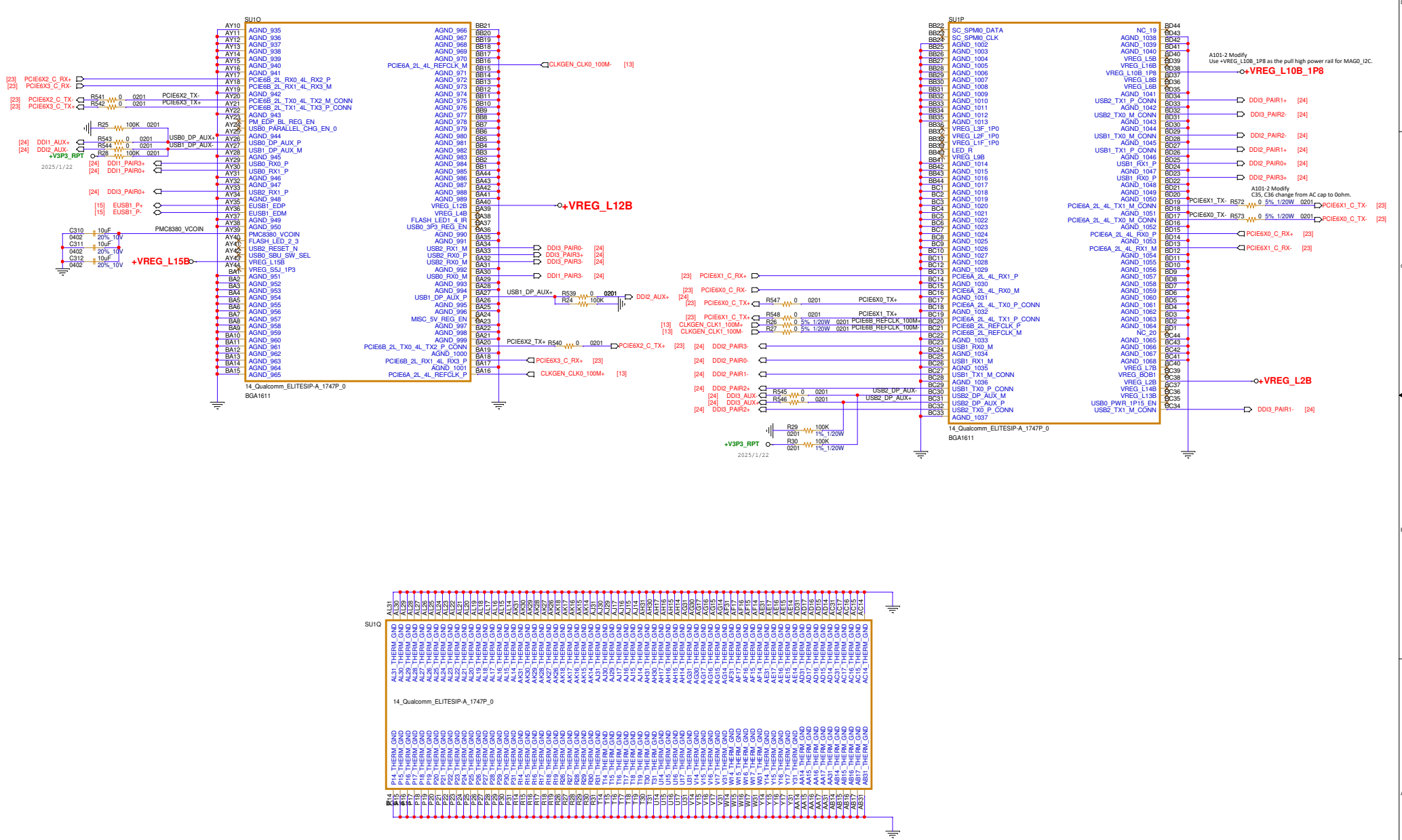
ADVANTECH		Classification Level	
		CONFIDENTIAL	
Title PG09: X ELITE SIP_801-1000			
Size B	PCB Name SOM-6820_A1		Rev 01-2
Date:	Tuesday, May 20, 2025		Sheet 9 of 31



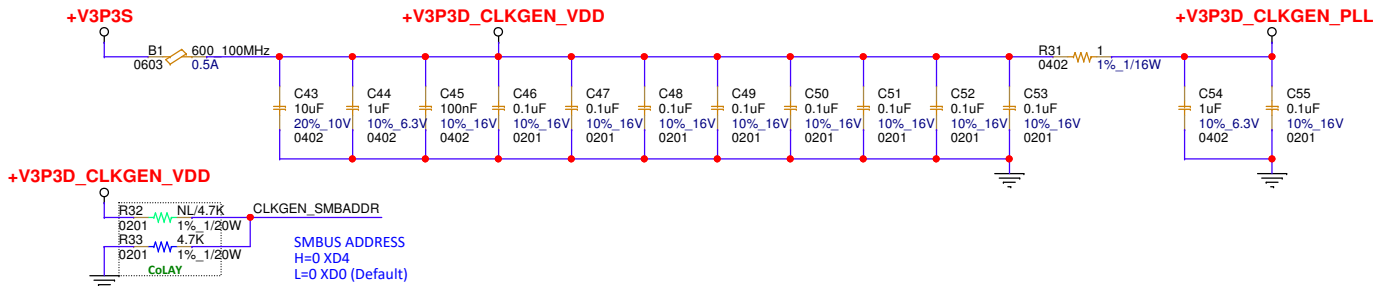
A101-2 Modify: Add CSI.



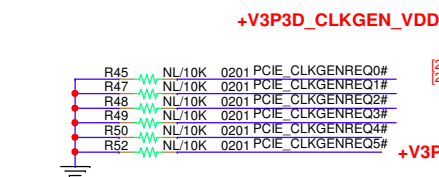
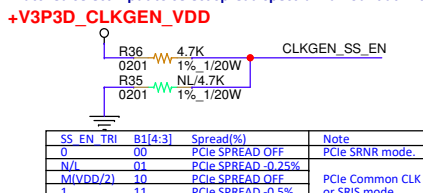




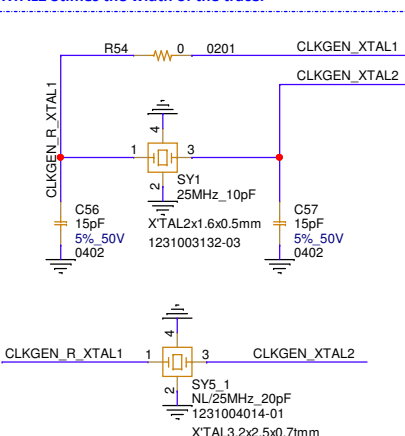
ExtCLK Gen for PCIe GEN 4



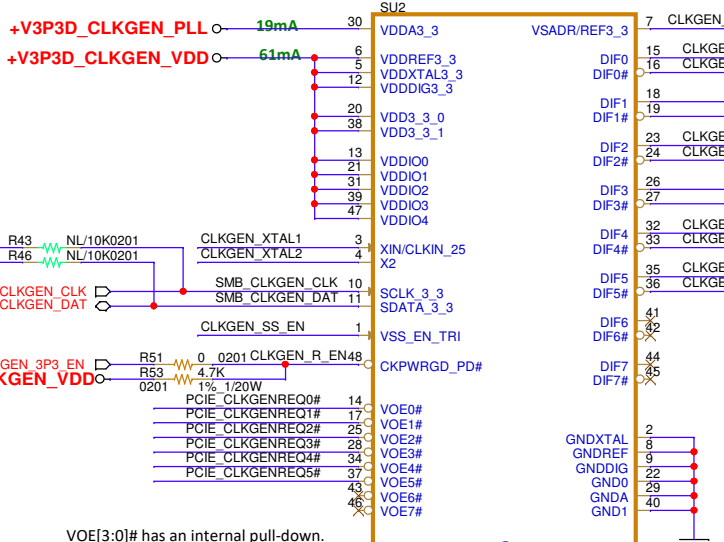
Latched select input to select spread spectrum amount at initial power up



Note: Separate net CLKGEN XTAL1 and XTAL2 3 times the width of the trace.

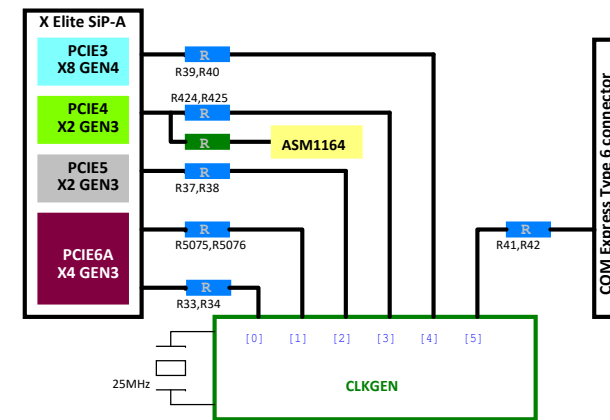


3225 and 2016 co-lay



VOE[3:0]# has an internal pull-down.

X Elite SiP Hamoia Lanes		PCIe								PEG							
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
		x4 GEN4 PCIe-6A				x2 GEN3 PCIe-5		x2 GEN3 PCIe-4		X8 GEN4 PCIe-3							
		x2 GEN4 PCIe-6A		x2 GEN4 PCIe-6B													
SOM-6820	OPTION1	X4 COMe #0-3				X2 COMe #4-5		X2 ASM1164		X8 COMe PEG#0-7							
	OPTION2	X2 COMe #0-1		X2 COMe #2-3				X2 COMe #6-7									

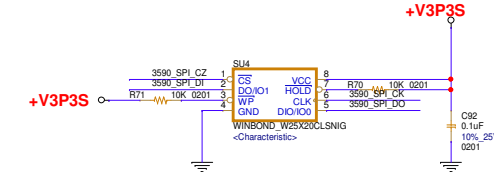
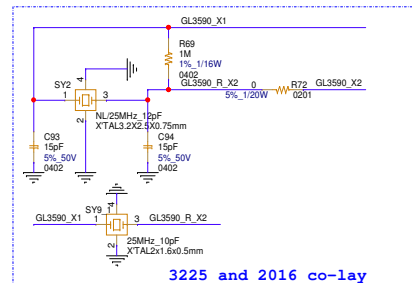
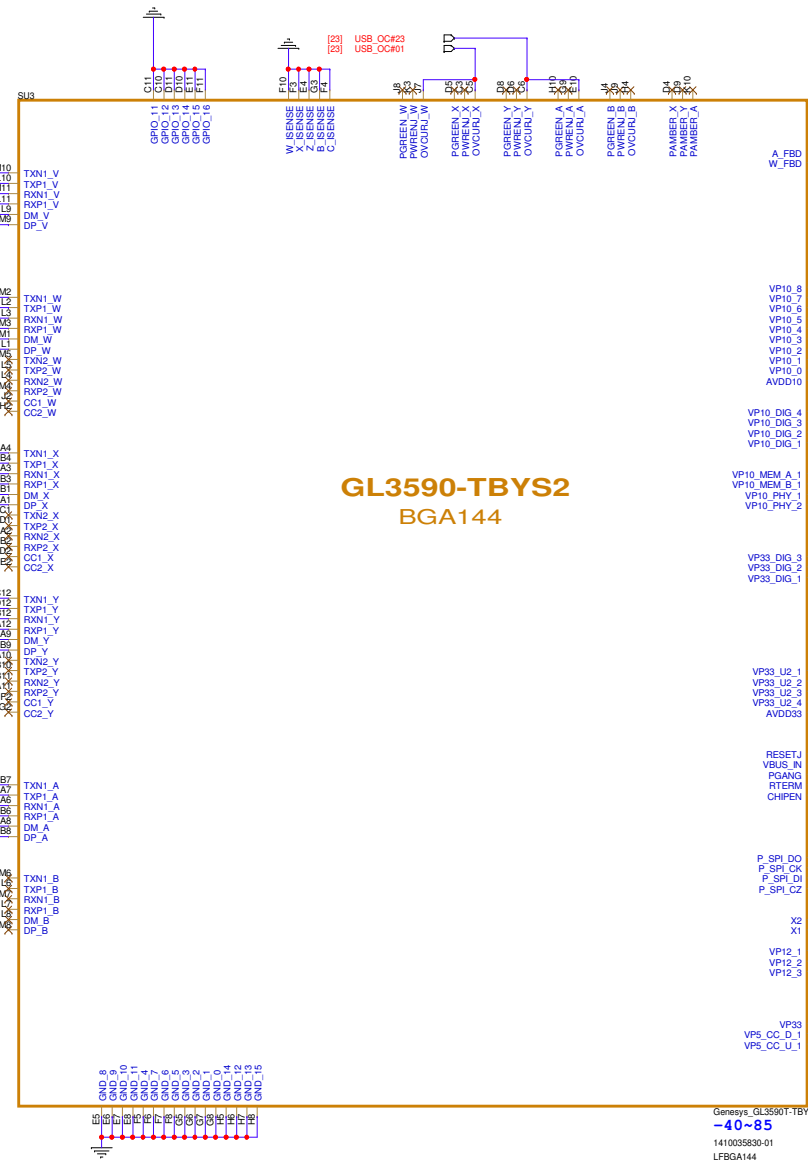
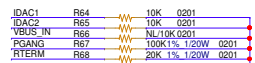
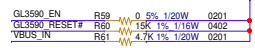
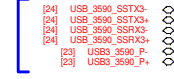
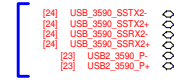
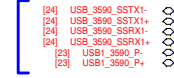
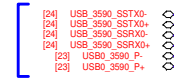
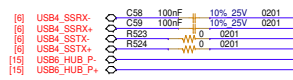


Note: 600ppm clock difference allowed between CPU and PCIe slot

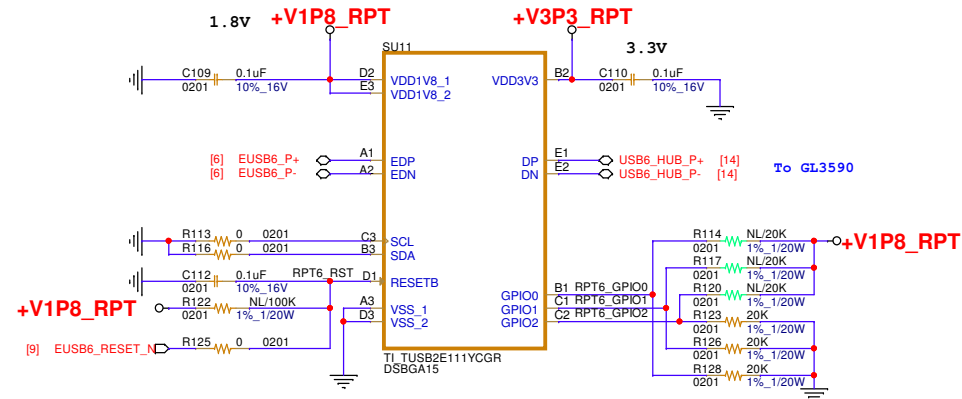
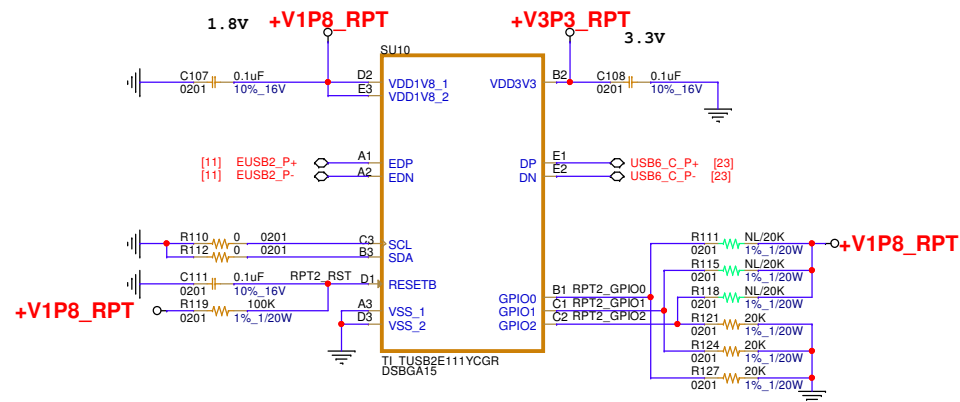
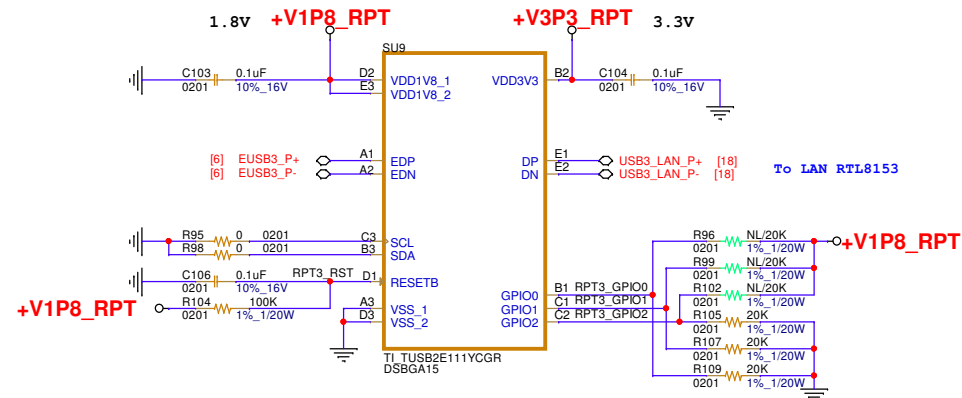
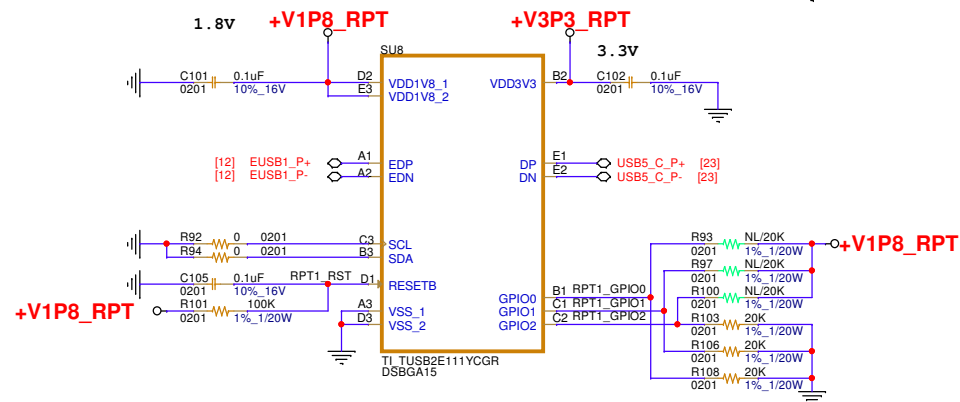
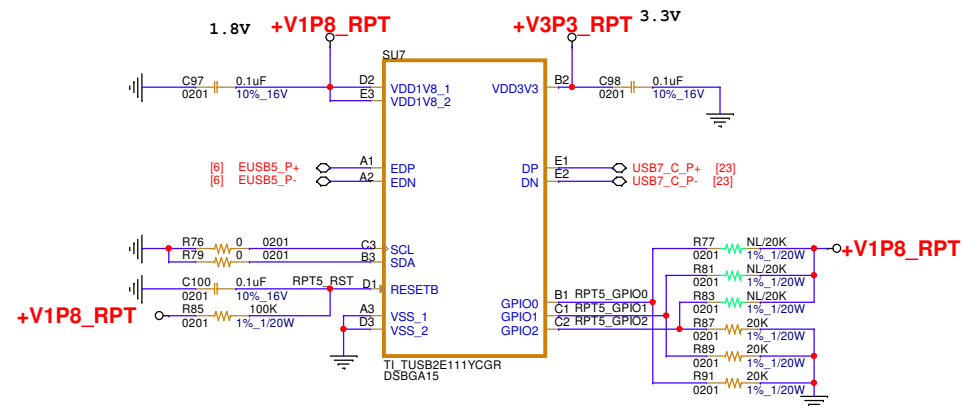
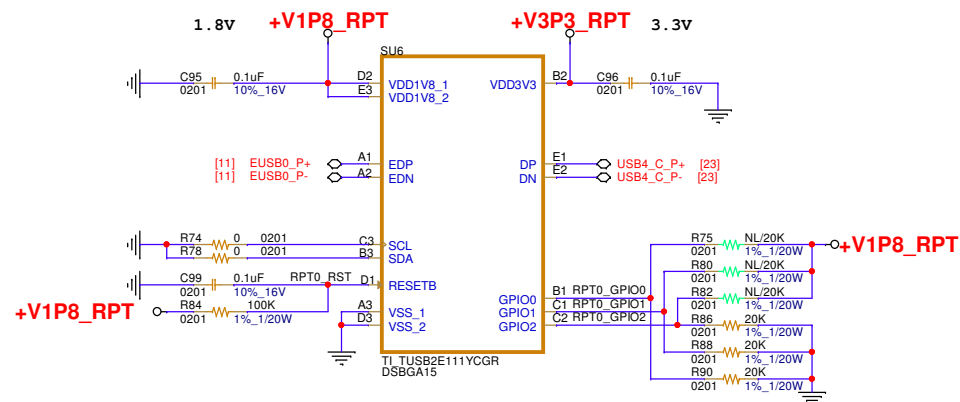
15	CLKGEN_CLK0_R_100M+	R37	0	0201	CLKGEN_CLK0_100M+	[12]	To PCIe6A for X4/X2 GEN3
16	CLKGEN_CLK0_R_100M-	R34	0	0201	CLKGEN_CLK0_100M-	[12]	
18	CLKGEN_CLK1_R_100M+	R38	0	0201	CLKGEN_CLK1_100M+	[12]	To PCIe6B for x2 GEN3
19	CLKGEN_CLK1_R_100M-	R39	0	0201	CLKGEN_CLK1_100M-	[12]	
23	CLKGEN_CLK2_R_100M+	R40	0	0201	CLKGEN_CLK2_100M+	[5]	To PCIe5 for x2 GEN3
24	CLKGEN_CLK2_R_100M-	R41	0	0201	CLKGEN_CLK2_100M-	[5]	
26	CLKGEN_CLK3_R_100M+	R42	0	0201	CLKGEN_CLK3_100M+	[6]	To PCIe4 for x2 GEN3 (OPTION)
27	CLKGEN_CLK3_R_100M-	R44	0	0201	CLKGEN_CLK3_100M-	[6]	
32	CLKGEN_CLK4_R_100M+	R40	0	0201	CLKGEN_CLK4_100M+	[6]	To PCIe3 for x8 GEN3
33	CLKGEN_CLK4_R_100M-	R41	0	0201	CLKGEN_CLK4_100M-	[6]	
35	CLKGEN_CLK5_R_100M+	R42	0	0201	CLKGEN_CLK5_100M+	[23]	To carrier board
36	CLKGEN_CLK5_R_100M-	R44	0	0201	CLKGEN_CLK5_100M-	[23]	

ADVANTECH		Classification Level		CONFIDENTIAL	
Title		PG13: CLKGEN_9FGL08		Rev D1-2	
Size B		PCB Name		SOM-6820_A1	
Date:		Tuesday, May 20, 2025		Sheet 13 of 31	

USB3 Hub



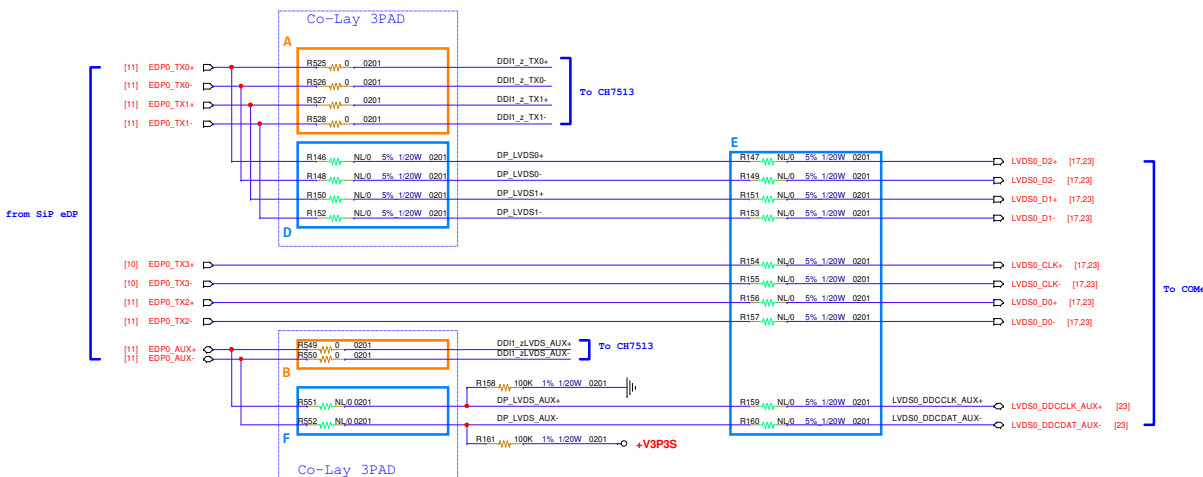
eUSB to USB2 Repeater



X Elite SIP					USB3_SS	USB4_SS				
	eUSB0	eUSB1	eUSB2	eUSB5	eUSB3	eUSB6				
					RTL8153	GL3590				
COM Express	USB2_P4	USB2_P5	USB2_P6	USB2_P7	GbE	USB3_P0	USB3_P1	USB3_P2	USB3_P3	

		Classification Level CONFIDENTIAL		CONFIDENTIAL DRAWINGS COVERED BY ADAPTIVE/TECH RECHARGEMENT SHALL MAINTAIN THE CONFIDENTIALITY AND SHALL BE LABELED FOR BREACH.
		Title PG15: eUSB to USB2 REPEATER		
Size A3	PCB Name SOM-6820_A1			Rev 01-2
Date:	Tuesday, May 20, 2025	Sheet	15 of	31

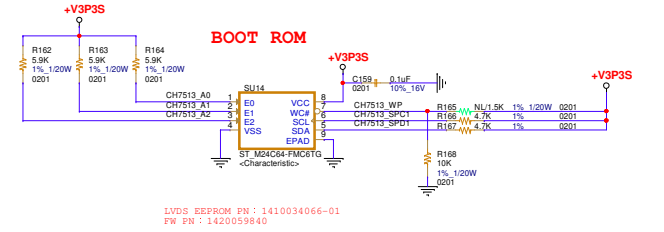
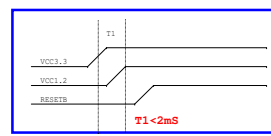
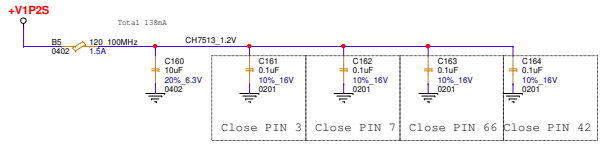
eDP To LVDS



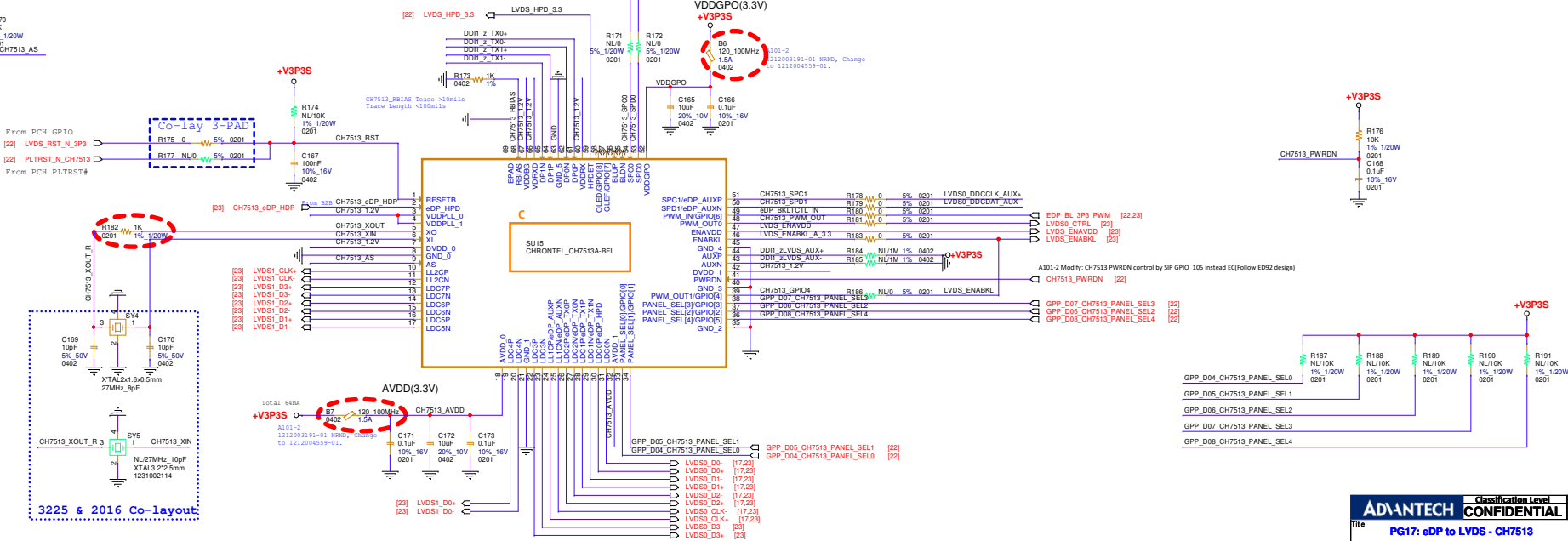
Copy from SOM-5885

LVDS / eDP BOM Option table						
	A	B	C	D	E	F
LVDS	V	V	V	X	X	X
eDP	X	X	X	V	V	V

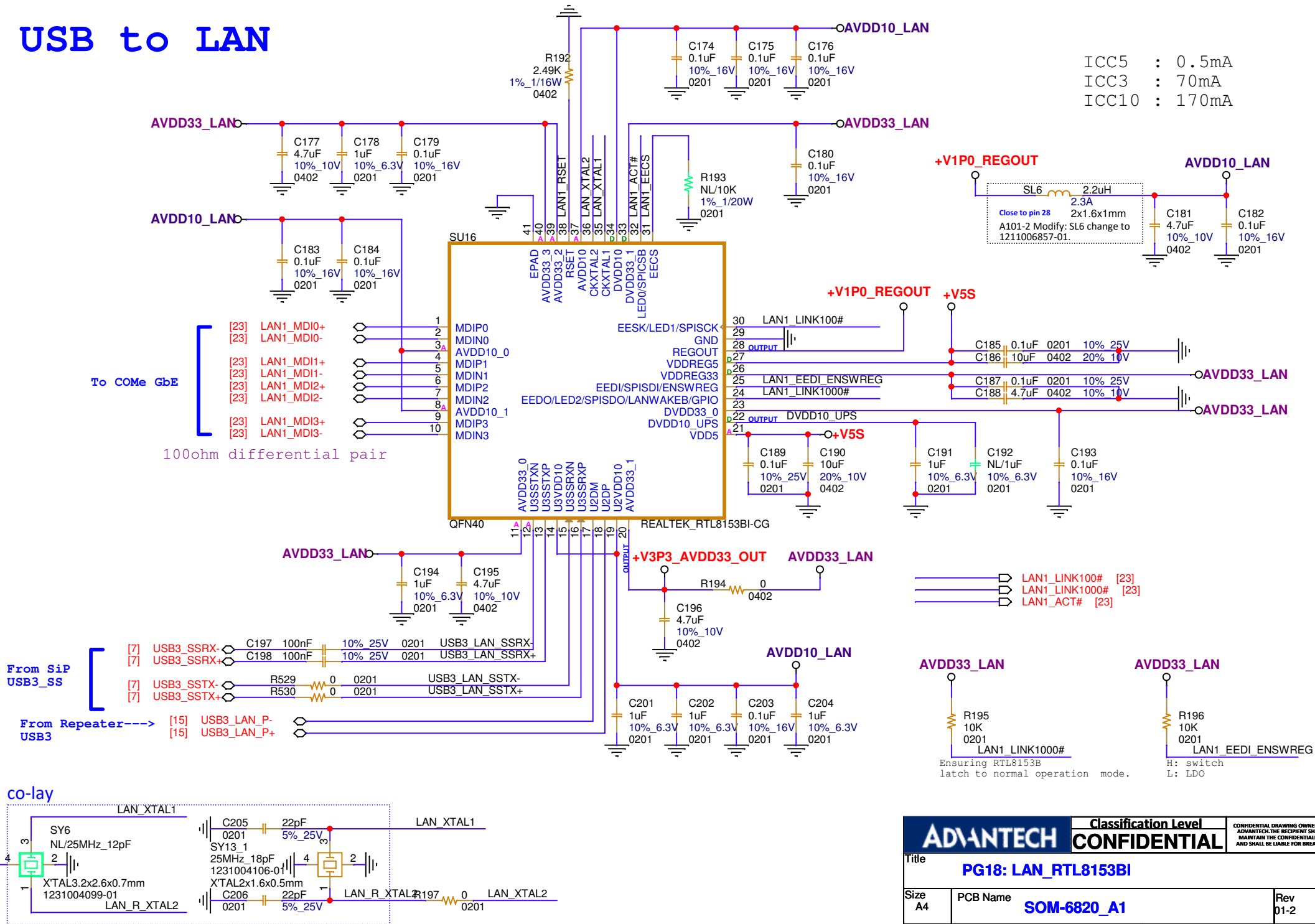
+V1P2S@ 0.2A



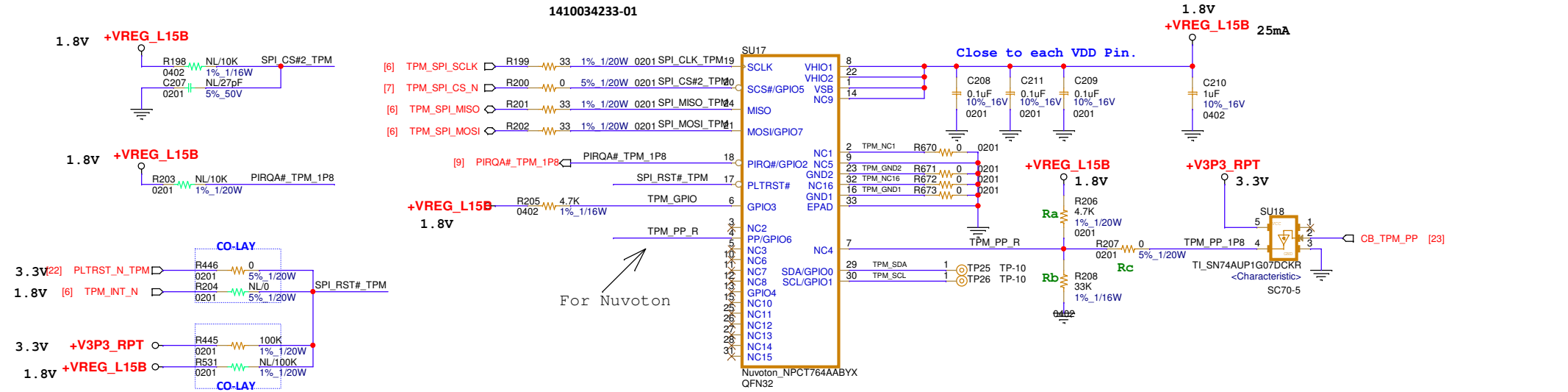
+V3P3S



USB to LAN



TPM Infineon SLB9672 / Nuvoton(NPCT76x) / 國民技術 (Z32H330x) / ST(ST33x) for Co-layout Common IP



Ra	Rb	Rc	Physical Presence special commands
NL	V	NL	Disable. *
V	NL	NL	Enable.
NL	V	V	Enable/Disable by Carrier Board.

* Default.

Co-Layout PN List: (Follow KCC).

Vendor	PN / MPN	Discription	OP Temp.	Verification	Note
Infineon	1410033322-11	IC SLB9672XU2.0 FW15.23 TPM 2.0 SPI UQFN 32P	-40 ~ 85 C	Verified	2023.11.07 Update
Infineon	SLB9672VU2.0 FW15.23	IC SLB9672VU2.0 FW15.23 TPM 2.0 SPI UQFN 32P	-20 ~ 85 C	TBD	
Nuvoton	1410033589-01	IC NPCT760AABYX TPM2.0 SPI FW7.2.3.1 070 QFN 32	0 ~ 70 C	Verified	
Nuvoton	1410034233-01	IC NPCT764AABYX TPM2.0 SPI FW7.2.3.1 -4085 QFN32	-40 ~ 85 C	TBD	
國民技術	Z32H330TC-SQN-2701	IC Z32H330TC-SQN-2701 TCM2.0 SPI QFN 32	-20 ~ 85 C	TBD	
ST	ST33HTPH2X32AHE4	IC ST33HTPH2X32AHE4 FW1.769 TPM 2.0 SPI UQFN 32P	-40 ~ 105 C	TBD	
ST	ST33KTPM2X32CKE2	IC ST33KTPM2X32CKE2 FW9.256 TPM 2.0 SPI UQFN 32P	-40 ~ 105 C	TBD	

For different OP Temp w/o Advantech PN, the MPN is listed above. EE can apply new PN for it.

ADVANTECH

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MAINTAIN THE CONFIDENTIALITY
AND SHALL BE LIABLE FOR BREACH.

Title

PG19: TPM - SLB9672A

Size
B

PCB Name
SOM-6820_A1

Rev
01-2

Date:

Tuesday, May 20, 2025

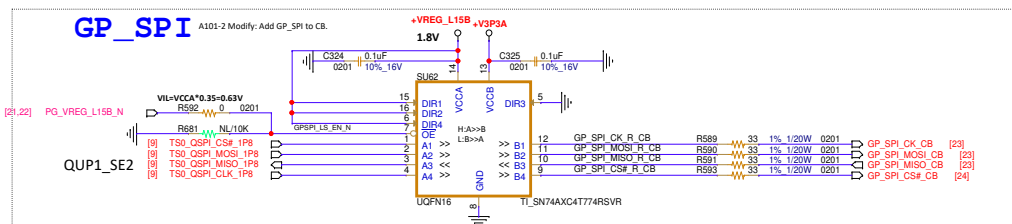
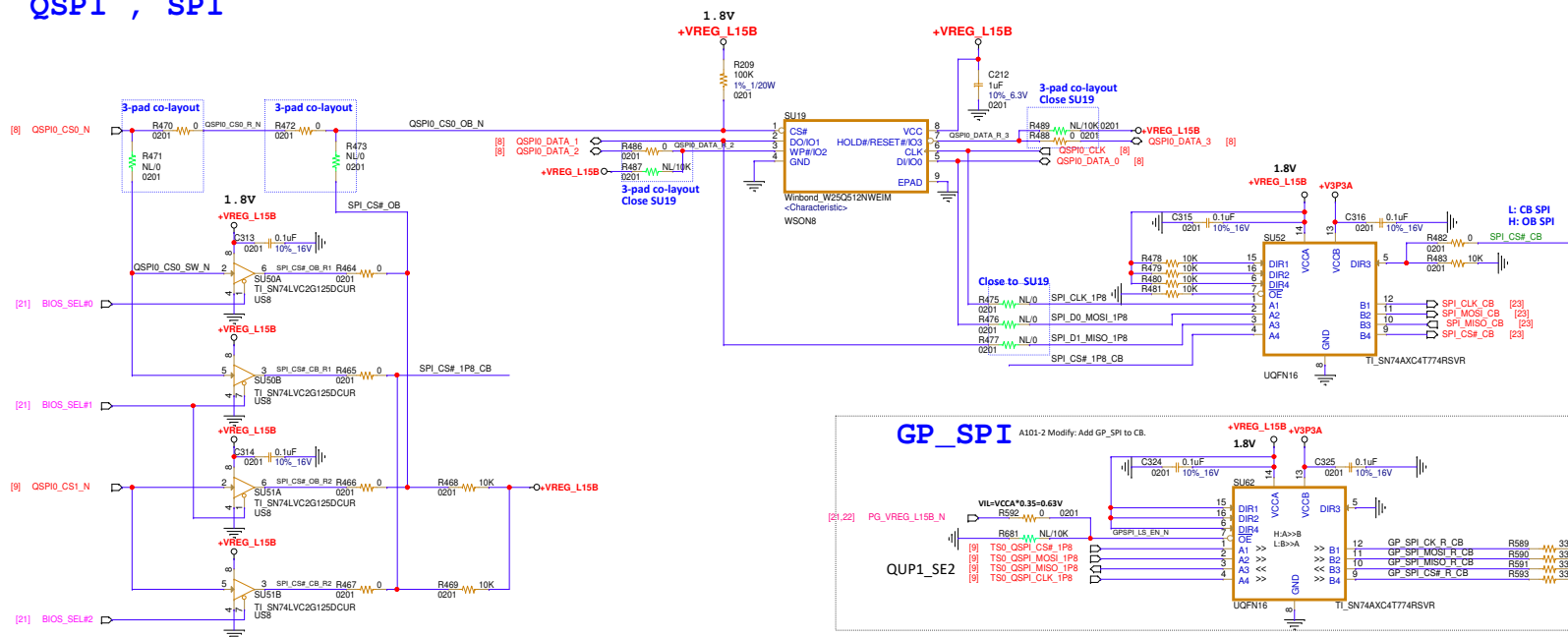
Sheet

19

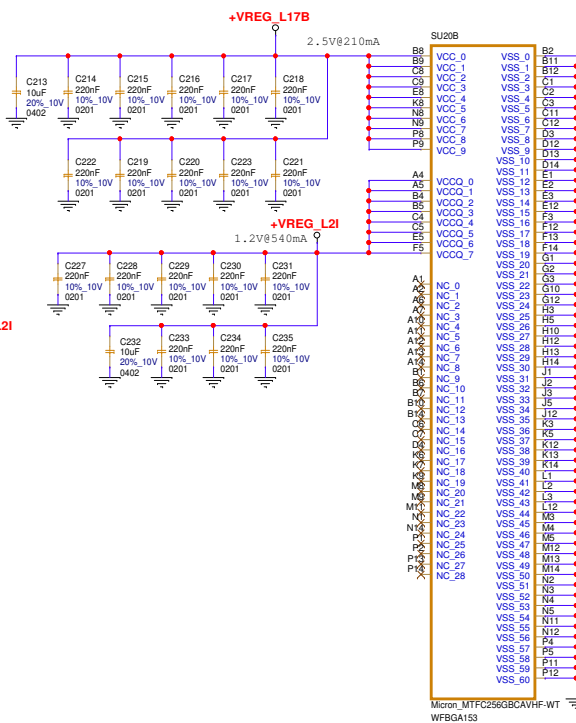
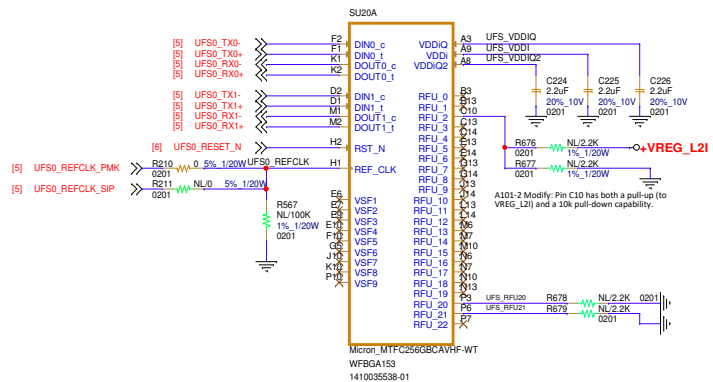
of

31

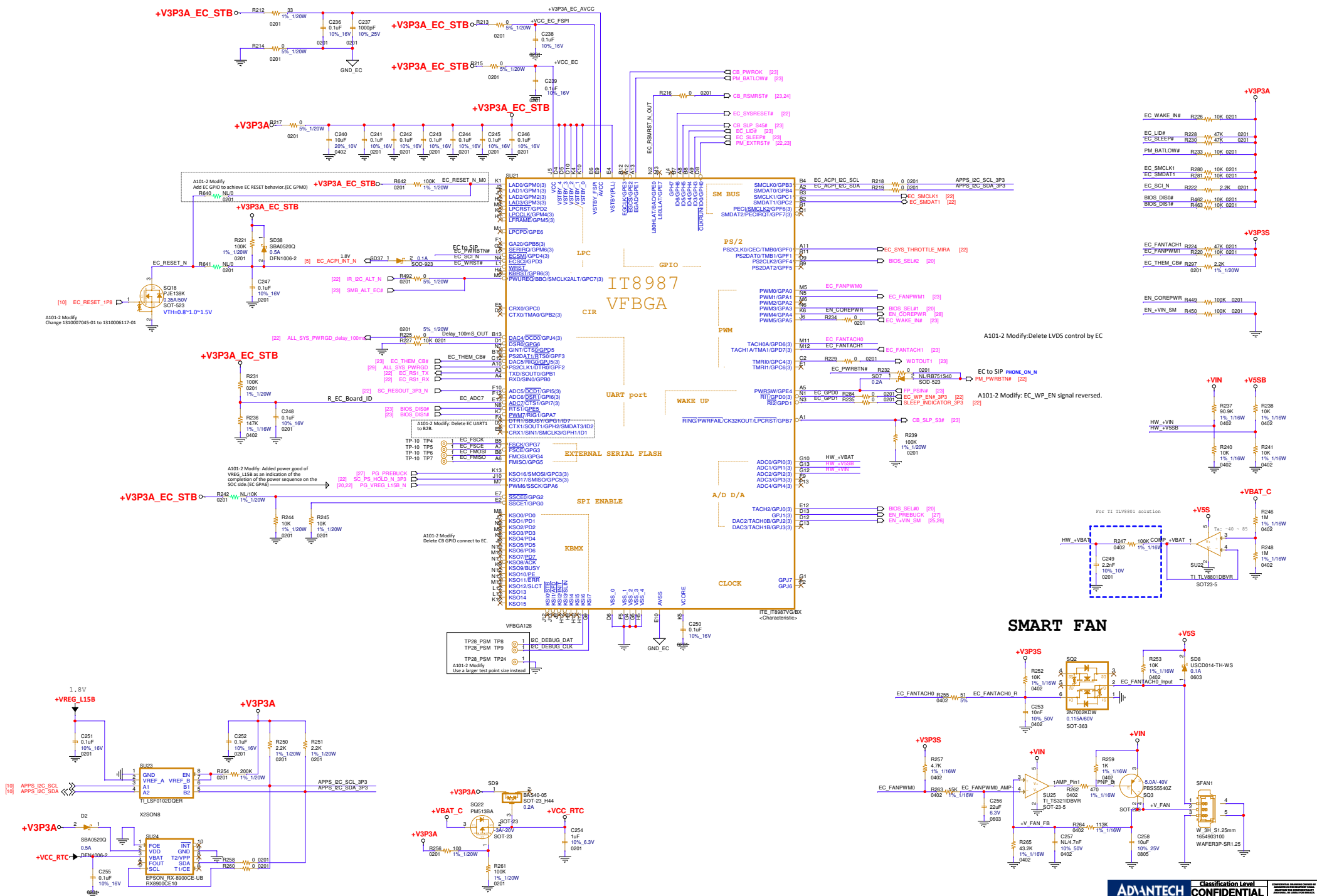
QSPI , SPI



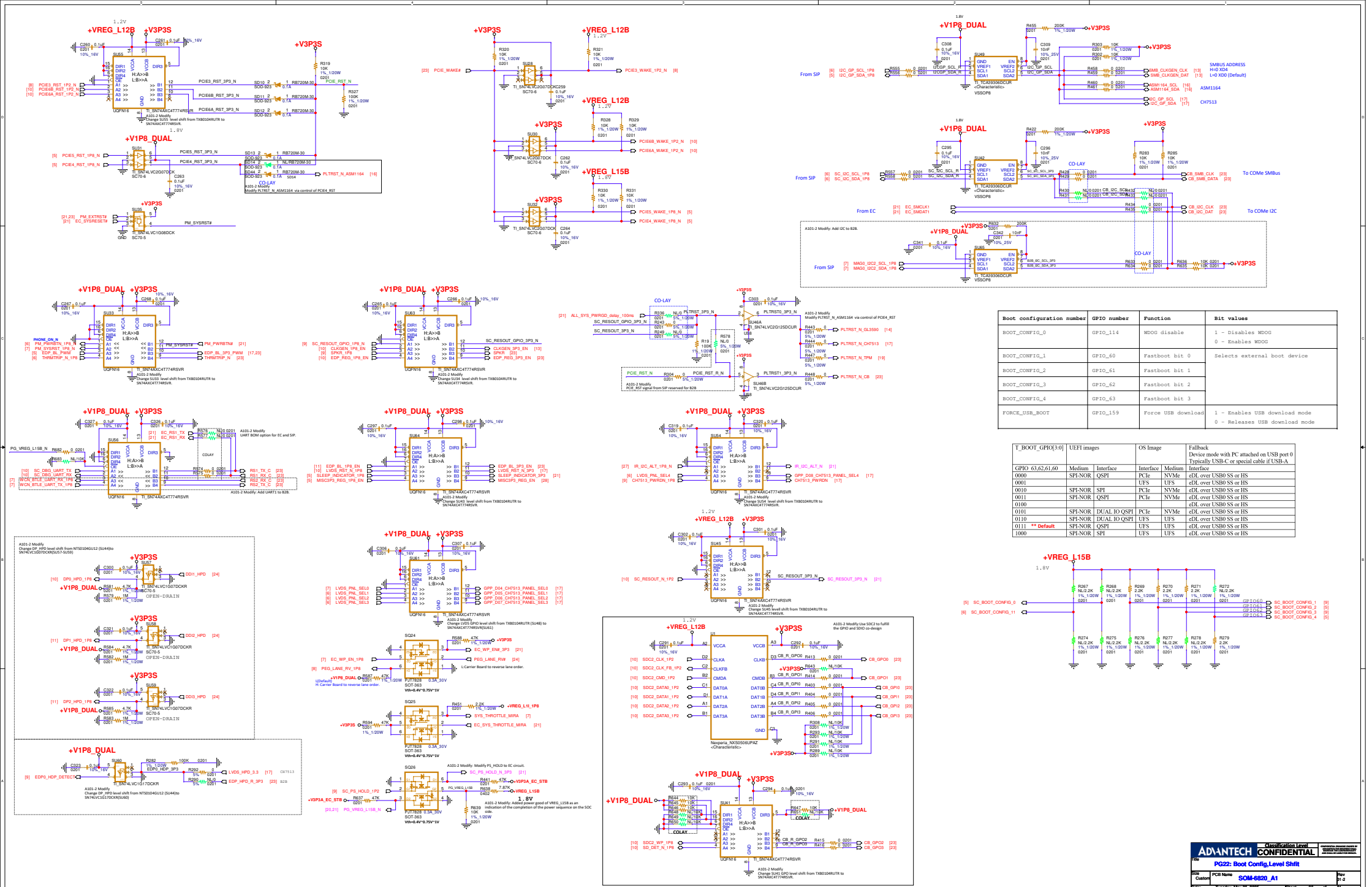
UFS

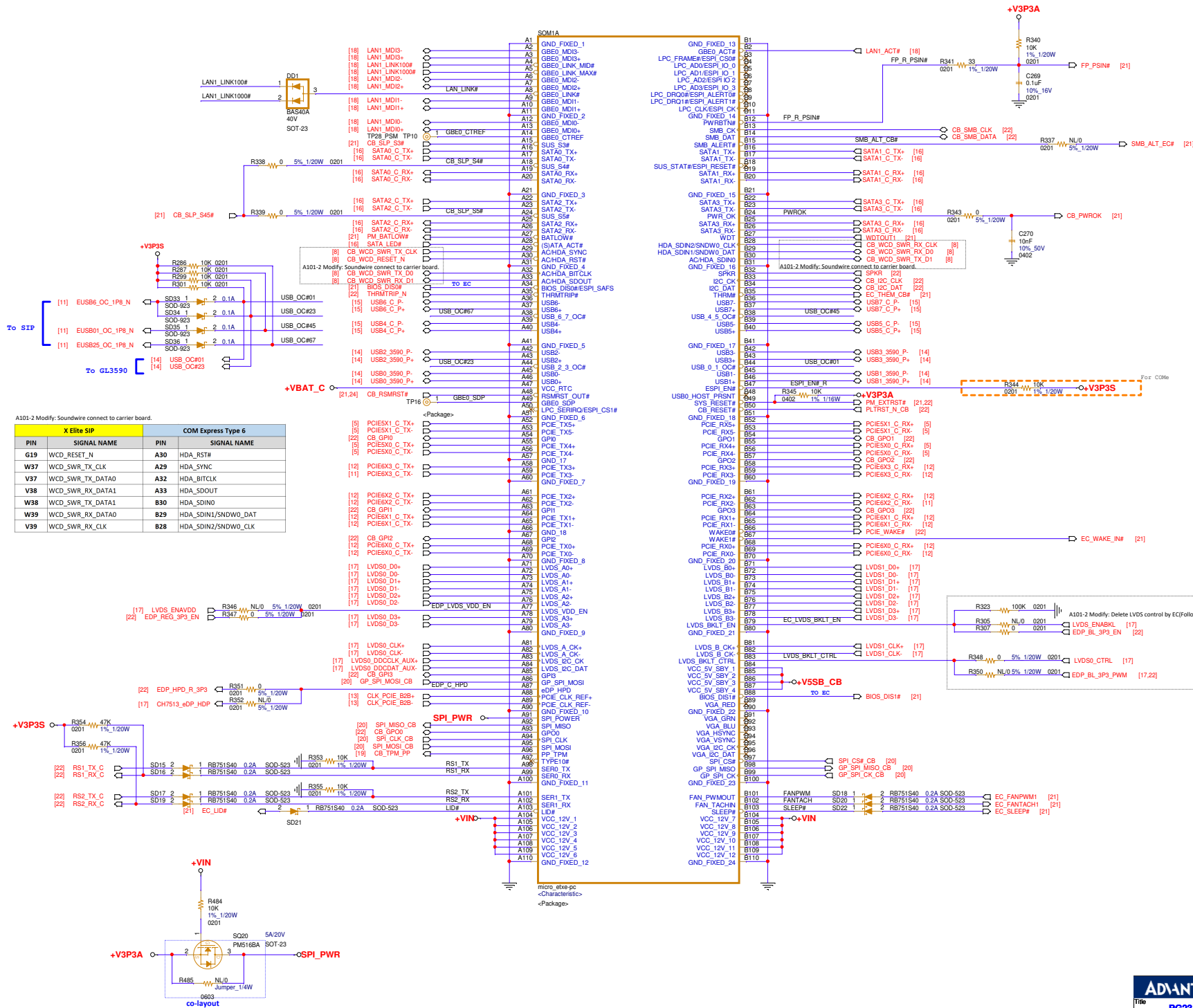


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Title PG20: QSPI NOR,UFS			
Size C	PCB Name SOM-6820_A1		Rev 01-2
Date:	Tuesday, May 20, 2025	Sheet	20 of 24



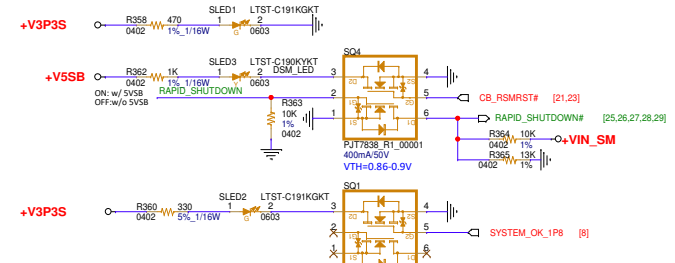
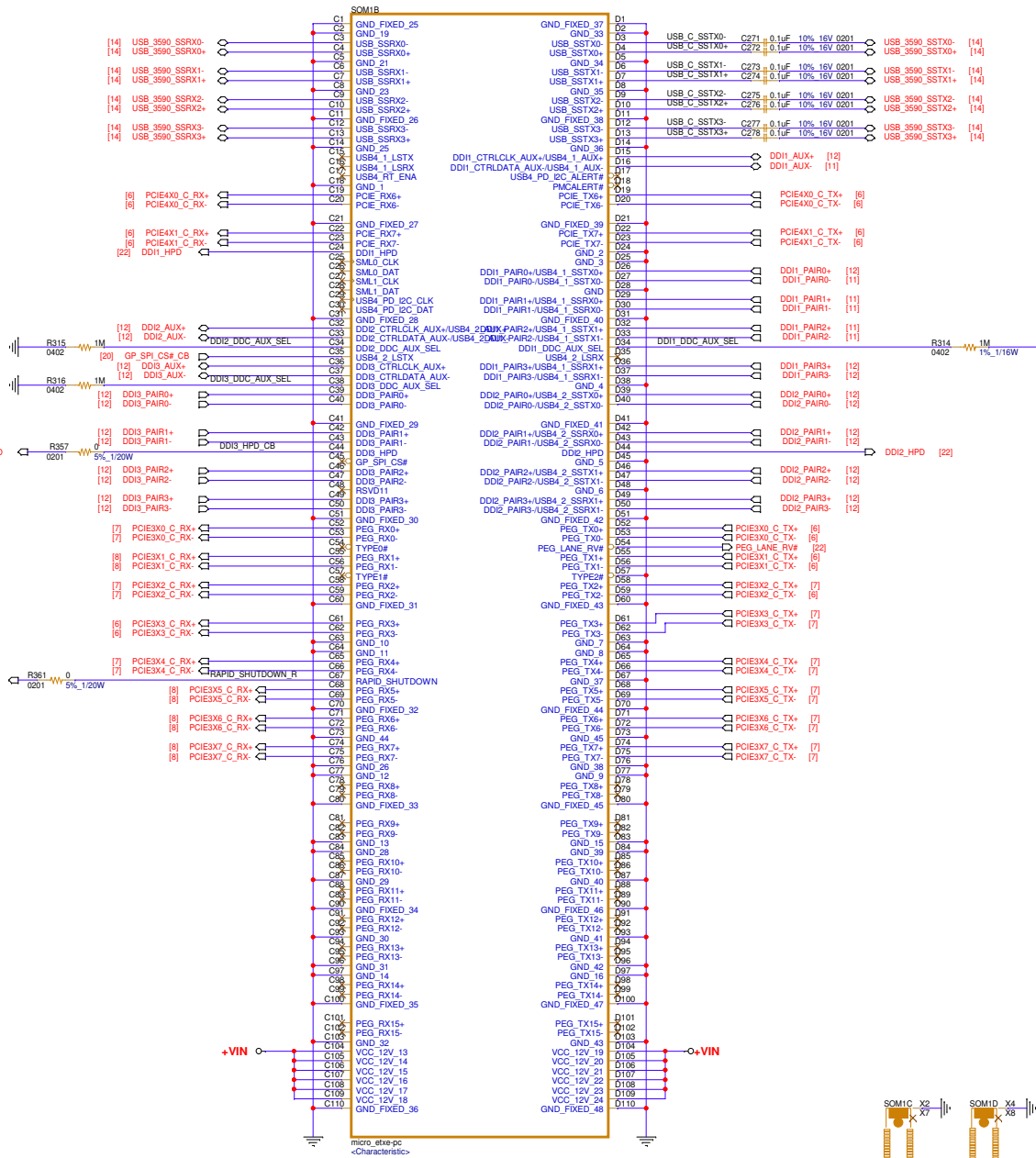
SMART FAN



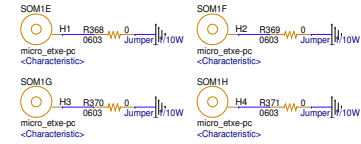
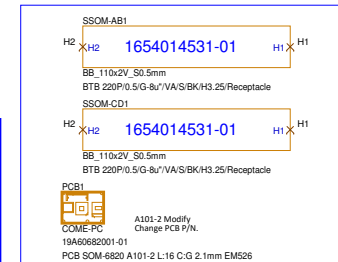
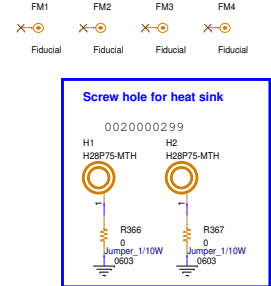


A101-2 Modify: Soundwire connect to carrier board.

X Elite SIP		COM Express Type 6	
PIN	SIGNAL NAME	PIN	SIGNAL NAME
G19	WCD_RESET_N	A30	HDA_RST#
W37	WCD_SWR_TX_CLK	A29	HDA_SYNC
V37	WCD_SWR_TX_DATA0	A32	HDA_BITCLK
V38	WCD_SWR_RX_DATA0	A33	HDA_SDOUT
V38	WCD_SWR_TX_DATA1	B30	HDA_SDINO
W39	WCD_SWR_RX_DATA0	B29	HDA_SDINI1/SDNDW0_DAT
V39	WCD_SWR_RX_CLK	B28	HDA_SDINI2/SDNDW0_CLK



For BOM 1654014531-01



**+VIN
(12V)**

Via X17
+VIN

Via X17
+VIN_SM @ 5A(4.75V@ 16.1A)

+VIN_SM

+V5SB_CB @ 3A

Via X3
+V5SB_CB

+V5SB @ 3A

Via X3
+V5SB

+VCC_OP

+V5SB

+VIN

+VIN_SM

+V3P3A

<VR Specification>
DC TOLERANCE=+/-3%
RIPPLE=+/-66mV
DC+AC TOLERANCE=+/-5%
I_MAX= 1.7A
OCP= 3.5A

+VIN_SM

VR_+V3P3A_VCC

+V3P3A

[24,26,27,28,29] RAPID_SHUTDOWN#

$$V_{out} = 0.6 * (1 + FR1/FR2) = 3.3029V$$

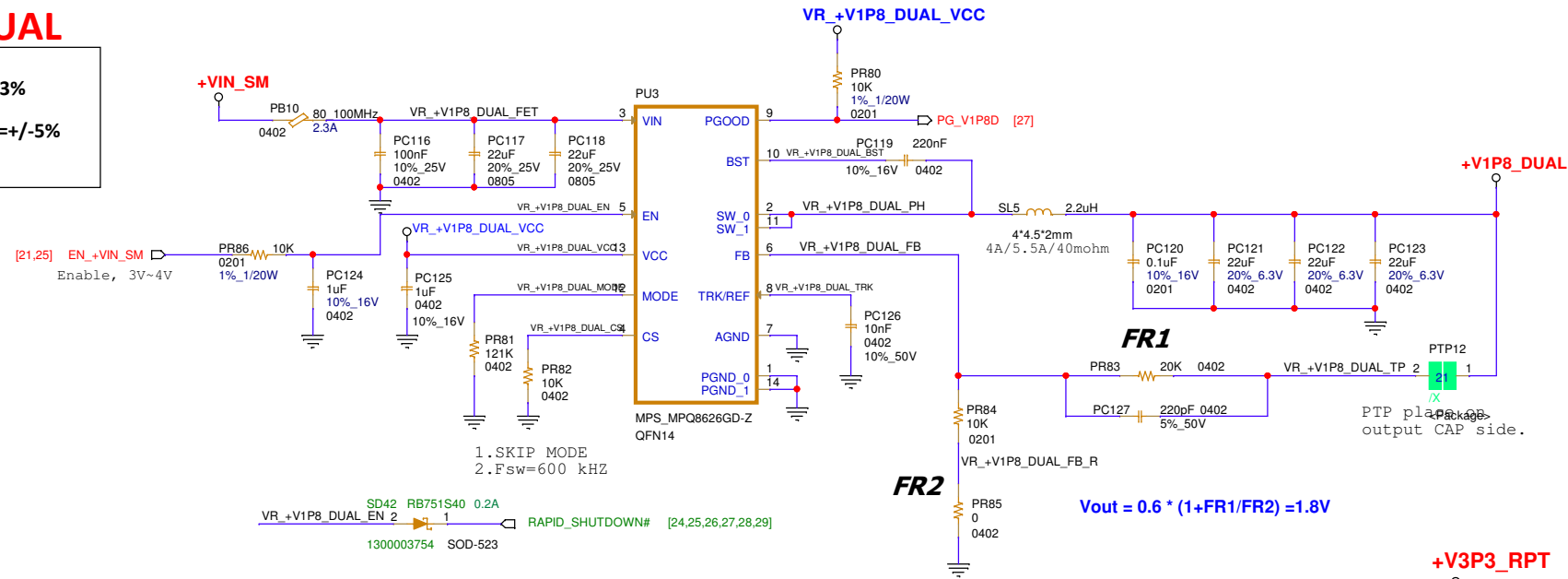
PTP place on output CAP side.

VR_+V3P3A_EN SD26 RB751S40 0.2A
1300003754 SOD-523

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Title				
PG25: DCIN,+V3P3A				
Size A3	PCB Name			
SOM-6820_A1		Rev 01-2		
Date: Tuesday, May 20, 2025		Sheet 25 of 31		

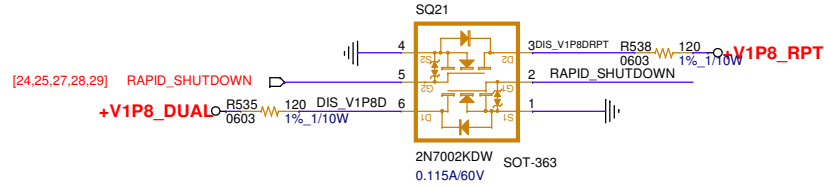
+V1P8_DUAL

<VR Specification>
DC TOLERANCE=+/-3%
RIPPLE=+/-36mV
DC+AC TOLERANCE=+/-5%
I_MAX= 2.5A
OCP= 3.5A



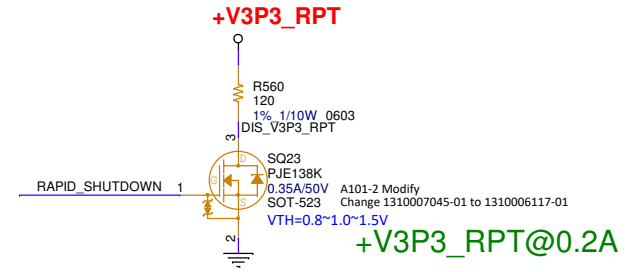
+V1P8_RPT

<VR Specification>
+V1P8_RPT
VOUT=1.8V
DC+AC TOLERANCE=+/-5%
I_MAX= 2.34A

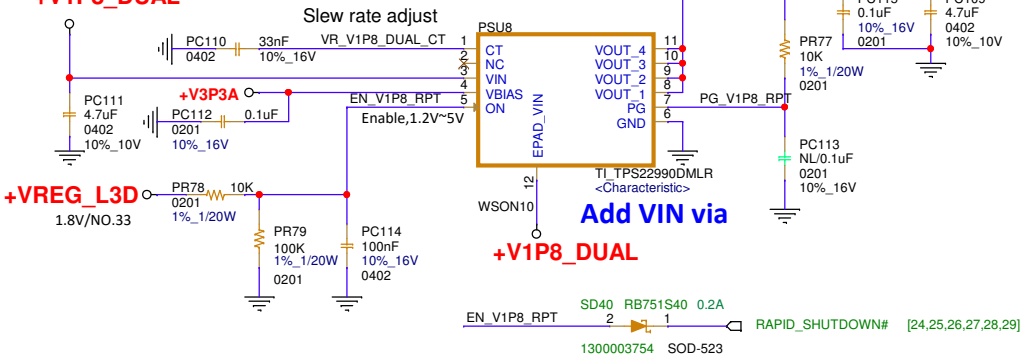


+V3P3_RPT

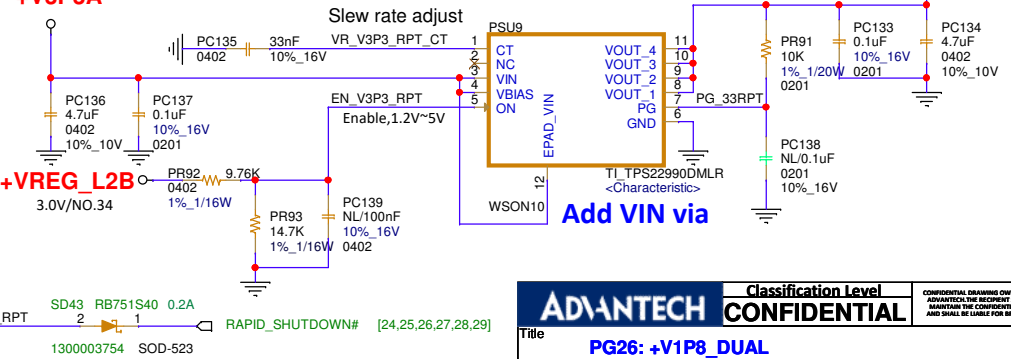
<VR Specification>
+V3.3S
VOUT=3.3V
DC+AC TOLERANCE=+/-5%
I_MAX= 2.34A



+V1P8_DUAL



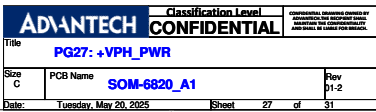
+V3P3A



ADVANTECH		Classification Level	CONFIDENTIAL
Title PG26: +V1P8_DUAL			
Size B	PCB Name SOM-6820_A1		Rev 01-2
Date: Tuesday, May 20, 2025	Sheet 26	of 31	

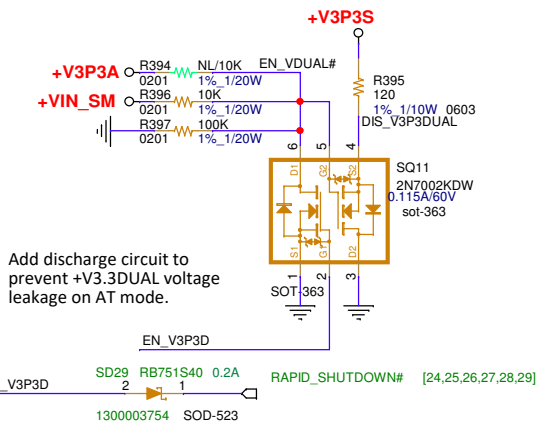
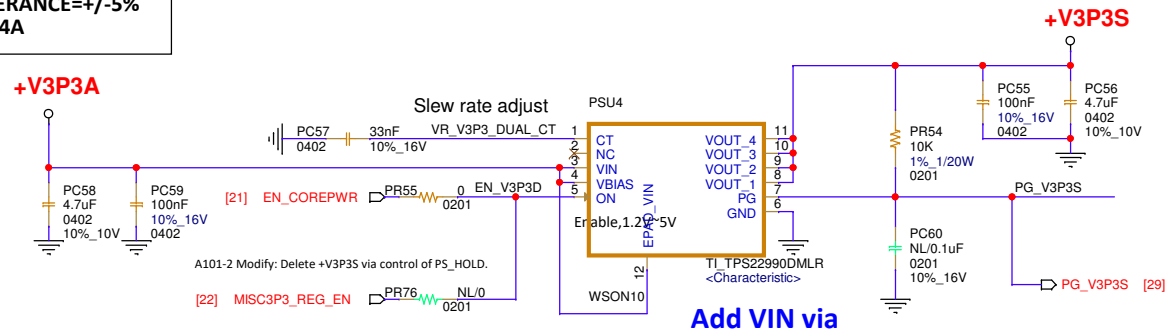
<VR Specification>
VOUT=3.3V
DC TOLERANCE=+/-2%
RIPPLE=+/-66mV
DC+AC TOLERANCE=+/-3%
I_TDC=20A
I_MAX= 50A
OCP= 60A(Fixed)

R _{ADDR} (kΩ)	Device Address (R _{IREF} = 60.4kΩ)	Device Address (R _{IREF} = 180kΩ)
4.99	30h	40
15	31h	41
24.9	32h	42
34.8	33h	43
45.3	34h	44
54.9	35h	45
64.9	36h	46
75	37h	47
200	3Fh	4F



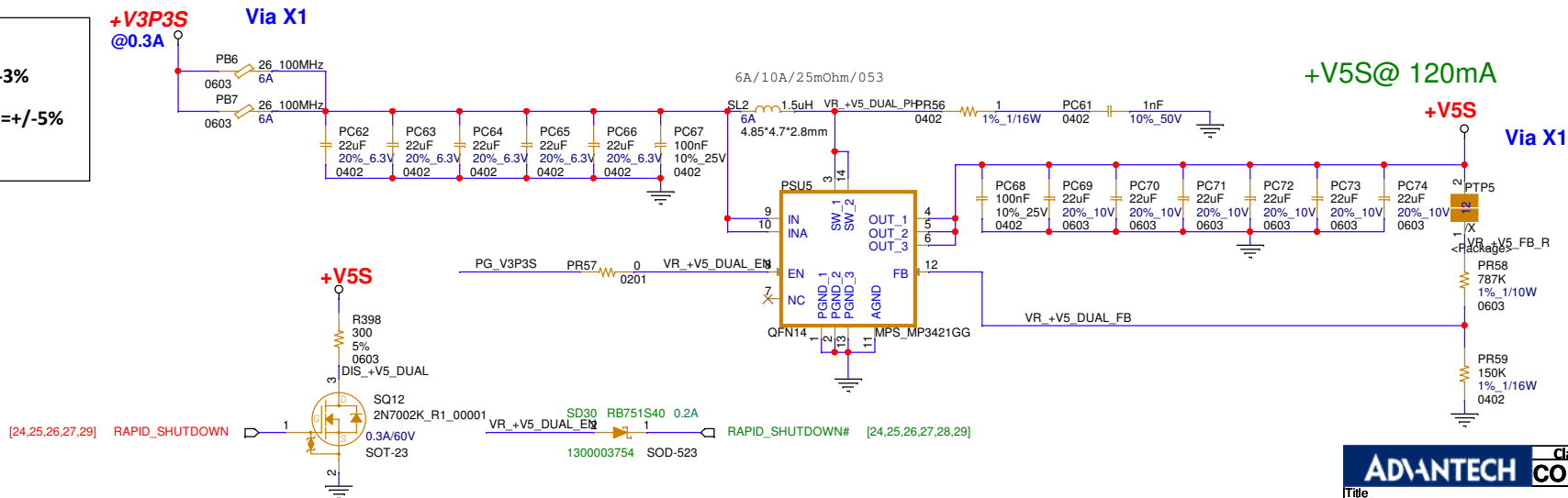
+V3P3S

<VR Specification>
+V3.3S
VOUT=3.3V
DC+AC TOLERANCE=+/-5%
I_MAX= 2.34A



+V5S

<VR Specification>
VOUT=5V
DC TOLERANCE=+/-3%
RIPPLE=+/-100MV
DC+AC TOLERANCE=+/-5%
I_MAX= 0.12A
OCP= 5A



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Title				
PG28: +V3P3S,+V5S				
Size B	PCB Name	SOM-6820_A1		Rev D1-2
Date:	Tuesday, May 20, 2025	Sheet	28	of 31

<VR Specification>
VOUT=1.2V
AC+DC TOLERANCE =+/-5%
I_MAX=1A
OCP=2.2A



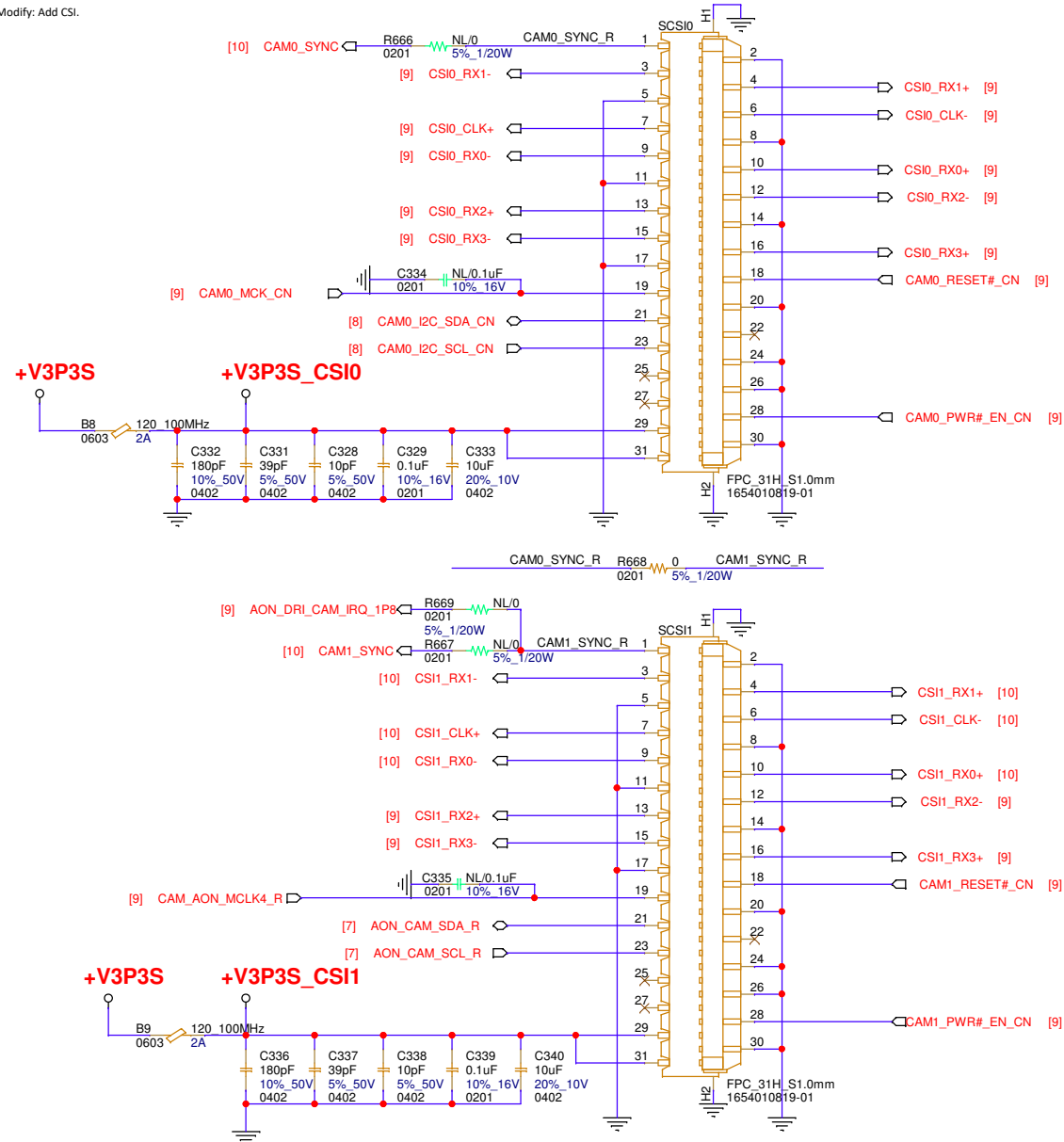
<VR Specification>
VOUT=1.05V
AC+DC TOLERANCE =+/-5%
I_MAX=0.6A
OCP=2.2A



CSIO, CSI1

A101-2 Modify: Add CSI.

Follow ROM-ED92 MIPI-CSI pin define.(20250514)



ADVANTECH		Classification Level	CONFIDENTIAL	CONFIDENTIAL DRAWING OWNED BY ADVANTECH. THE RECIPIENT SHALL MAINTAIN THE CONFIDENTIALITY AND SHALL BE LIABLE FOR BREACH.
Title				
PG30: MIPI-CSI HDR				
Size B	PCB Name		SOM-6820_A1	Rev 01-2
Date:	Tuesday, May 20, 2025		Sheet 30 of 31	

SOM-6820 Revision History

SOM-6820 A101-1 PCB:19A60682000-01, 96 BOM:96966820000 Date: 2025/1

SOM-6820 A101-2 PCB:19A60682001-01, 96 BOM:96966820000 Date: 2025/5 (EVT2)

Page 5,6,10:internal pull-ups have been removed.(20250408 QC)(TS_I2C , IR_I2C,APPS_I2C)
Page 7,30:Add CSI0, CSI1
Page 7,22:Add I2C to B2B.
Page 8:SYSTEM_OK_1P8 change to GPIO_142
Page 8:Change the SoundWire connector SCN1 to SNDW1 and modify the SoundWire circuit.
Page 8,23:Soundwire connect to carrier board.
Page 9,20:Add GP_SPI to CB.
Page 9,17,21,23:Delete LVDS control by EC(Follow ED92 design).CH7513 PWRDN control by SIP GPIO_105 instead EC.
Page 10,22:SIP Debug UART to B2B, BOM option for EC .
Page 10,22:Use SDC2 to fulfill the GPIO and SDIO co-design.
Page 12:PCIE6X1 AC cap C35, C36 change to 0ohm.
Page 12:Use +VREG_L10B_1P8 as the pull high power rail for MAG0_I2C.
Page 18:SL6 change to 1211006857-01.
Page 21:SQ18 change to ummounted by default.
Page 21:Delete CB GPIO connect to EC
Page 21:Delete EC UART1 to B2B.
Page 21:EC_WP_EN signal reversed.
Page 21: Add EC GPIO to achieve EC RESET behavior.(EC GPM0)
Page 22:PCIE_RST signal from SIP reserved for B2B
Page 22:Change DP_HPD level shift from NTS0104GU12 (SU44)to SN74LVC1G07DCKR(SU57-SU59)
Page 22:Change eDP_HPD level shift from NTS0104GU12 (SU44)to SN74LVC1G17DCKR(SU60)
Page 22:Change Level shift to SN74AXC4T774RSVR (SU61,SU41,SU40,SU55,SU45,SU63,SU33,SU64,SU54)
Page 22:Modify PS_HOLD to EC circuit.(SQ26)
Page 22:Modify PLTRST_N_ASM1164 via control of PCIE4_RST
Page 22:Added power good of VREG_L15B as an indication of the completion of the power sequence on the SOC side.
Page 24:Change PCB P/N.
Page 27:PR38 connect to Remote VSENSE_M not GND.
Page 28:Delete +V3P3S via control of PS_HOLD
Change 1310007045-01 to 1310006117-01 (SQ18,SQ23,SQ17)